Differential Punishment and the Consequences for Labor Supply

Katherine B. Coffman and Dalila Figueiredo

October 30, 2026

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Abstract

Punishment is common at work and often discretionary, leaving room for bias and unintended effects on labor supply. We study both sides of this decision in a synchronous online experiment, pairing managers and workers in a real-effort project. We randomly vary whether the manager's sanction is implemented and whether the manager observes the worker's effort or only the project outcome. On average, managers do not punish Black and white workers differently. However, a racial gap appears when effort is observed and is low: at zero effort, managers are 16.5 percentage points more likely to sanction a Black worker and, on average, sanction them 1.3 tokens more. This gap is concentrated among men and closes near the average effort level. Evidence suggests that managers judge the same low-effort signal differently by workers' race. On the worker side, the threat of punishment leaves effort unchanged, but receiving a sanction reduces subsequent effort, with white men responding most strongly, and makes workers, women in particular, less willing to keep working for the same manager. In our setting, punishment lowers joint surplus, with the worker bearing the entire loss.

JEL: J15, J16, J71

1 Introduction

Punishment is a feature of many educational and professional environments. Teachers discipline students, supervisors sanction employees, and firms penalize misconduct. While punishment can incentivize effort and rule-following, it can also demotivate workers, ending up being costly to the organization. Furthermore, because these decisions are often discretionary, they may be subject to bias and discrimination.

Indeed, a growing body of field evidence suggests that punishment is not always levied equally across groups. Women often face harsher consequences than men following similar negative outcomes, as seen among financial advisors and surgeons (Egan et al., 2019; Sarsons, 2017), and female leaders receive more scrutiny and are terminated more often after poor firm performance (Carvalho, 2025; Ferrario and Carvalho, 2026). Racial disparities have also been documented. For example, Black students are substantially more likely to be expelled or suspended (Kinsler, 2011; Darling-Hammond et al., 2023), dark-skin football players more likely to receive foul calls (Magistro and Wack, 2023), and non-White employees face more disciplinary procedures (NHS England and NHS Improvement, 2020). These patterns matter not only because punishment directly affects the workers who receive it, but also because discipline can shape how an individual engages with the organization going forward, contributing to decisions to disengage or exit.

This existing literature raises several important questions. First, what causes unequal punishment? Do managers' interpretations of the same outcome vary by worker identity, and if so, is this due to biased beliefs about workers' effort? Field data rarely speak to managers' information, beliefs, or counterfactual punishment decisions, making it challenging to pin down these potential mechanisms. Second, does the threat or experience of punishment affect workers' labor supply, such as how hard they work or whether they stay in the organization? Finally, do these responses vary by worker race and gender? Understanding the full set of consequences of punishment requires studying both the manager's decision to punish and the worker's response to that decision.

To answer these questions, we conduct a large, synchronous experiment with more than 3,000 online participants. Individuals are randomly assigned to the role of the worker or manager, and paired. The worker must complete a real effort task (slider task), where their effort increases the chance that a joint project succeeds. If the outcome is good, both the manager and the worker receive 10 tokens. If the project fails and the outcome is bad, only the worker gets paid the same 10 tokens. Therefore, the worker has a financial incentive to

shirk, and the only two reasons not to are altruism and fear of punishment. Pairs play two rounds, and workers can then choose to continue into a third round with the same manager. Managers and workers learn the race and gender of their partners through an avatar that represents them and is visible throughout the study (Ruebeck, 2025).[1]. Participants build their avatar before learning the game. The gender of this avatar is fixed by the participant's stated gender in the demographics survey, and the skin-tone of the avatar corresponds to the stated race for more than 90% of participants. For power concerns, all managers are white, while workers are Black or white.

The game randomly implements two features: whether the manager can punish the worker by destroying a chosen amount of the worker's tokens, and whether the manager observes the worker's effort or only the final project outcome. Punishment costs the manager nothing but is socially costly, as any sanctioned tokens are simply destroyed. Workers know from the start whether they can be punished. By contrast, managers start the game knowing there is a 50% chance their sanctions will be implemented. In the first round, after the project's outcome is revealed, all managers decide on their sanctions. Managers learn whether their sanction is imposed (Punishment arm) or not (No-Punishment arm). If it is, they can punish in the following rounds. Otherwise, they know they will not have another opportunity to punish. Workers learn punishment immediately after the outcome is revealed and before the next round begins. In the No-Punishment arm, workers never hear about sanctions. The second treatment variation is whether the manager perfectly observes effort. In the Hidden-Effort treatment, managers observe only the project outcome (good or bad), whereas in the Known-Effort treatment they learn the worker's effort. In the first round, workers know that the manager will observe their effort with a 50% chance. At the end of the first round, workers learn whether the manager observed their effort and know the same will happen in the following rounds.

This design allows us to study whom managers punish, and under which conditions.

[1]We use the package developed by Ruebeck (2025) as the foundation of our avatar construction, making only minor changes to suit our setting.

Furthermore, by studying the two sides of the interaction together, we can also examine how punishment or its anticipation affects effort supply and the worker's willingness to continue the relationship.

To answer these questions, we conduct a controlled experiment that exogenously varies the possibility of punishment, while holding fixed the underlying production environment and controlling the information available to managers and workers. We compare otherwise similar manager–worker relationships in which punishment is or is not available, and we track effort, outcomes, and punishment decisions over repeated interactions. By studying the two sides of the interaction together, we can examine not only how punishment is allocated, but also how punishment affects subsequent effort supply and the continuation of the relationship. In addition, we exogenously vary the ability of the manager to perfectly monitor worker effort, allowing us to understand what role beliefs about effort play in driving differential punishment decisions.

We gather data from a large, synchronous experiment with more than 3,000 online participants. After an initial survey of demographics and economic preferences, participants are matched into manager-worker pairs for multiple rounds of a 'joint project', the success of which depends upon how much real effort the worker chooses to exert. A successful joint project pays both parties 10 tokens, while an unsuccessful one pays the worker 10 tokens but the manager nothing. Thus, workers have a direct financial incentive to shirk, but may choose to exert effort because of other-regarding preferences or to avoid punishment. The outcome of the joint project is revealed to both parties at the end of each round. We cross this treatment with variation in whether the manager perfectly observes effort. In the Hidden-Effort treatment, managers observe only the outcome of the project; in the Known-Effort treatment, they also see the effort the worker exerted.

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avatar is fixed by the participant's stated gender in the demographics survey. All participants then choose their avatar's skin tone, hair color, hairstyle, facial hair, and whether to wear glasses. To maximize statistical power to identify differences in levied punishment by worker race, we over-sample Black participants and assign them to the worker role, while white participants are randomly-assigned to either the manager or worker role. As a result, half of our study pairs feature a white manager with a Black worker, while the other half feature a white manager with a white worker.

Overall punishment rates are somewhat low. Managers punish in fewer than 1/3 of Round 1 decisions; conditional on punishing, the average sanction is just over 5 tokens (of a possible 10). Unsurprisingly, punishment is most common after negative outcomes. There is no gender gap in punishment received. Unconditionally, Black workers receive somewhat larger sanctions than white workers, but this largely reflects differences in outcomes. Conditional on the project's success and worker effort, we find no significant race or gender gap in punishment, on average.

We find that managers use punishment differently when they can observe effort. Managers who observe effort are more likely to punish, and impose larger sanctions, on average. As expected, punishment decreases as worker effort increases, consistent with managers punishing based upon perceived responsibility for the outcome.

Importantly, a racial gap emerges when effort is revealed. Among lower effort workers, Black workers receive larger sanctions than white workers. At no effort, managers are 16.5pp ($p < 0.05$) more likely to punish Black workers. Additionally, while the estimates are imprecise, our data suggests managers are approximately 8pp ($p \approx 0.15$) more likely to sanction a Black worker the maximum possible amount (10 tokens). The racial gap narrows as effort increases, closing at roughly the sample average effort. The gap operates mainly on the extensive margin, on whether to punish at all, and it is concentrated among male workers. We find no analogous gender gap at any effort level. The evidence therefore does not point to a broad tendency to punish Black workers more harshly. Rather, differential punishment

emerges in a specific part of the performance distribution: when the manager knows that the worker provided relatively little effort. Note that this pattern seems inconsistent with statistical or beliefs-based discrimination, which would predict a larger racial gap when effort is unobserved. Similarly, we find no evidence that managers expect less effort or hold higher standards for Black workers. Instead, our results suggest that the same, precise low-effort signal is punished differently by race, even when stated beliefs are similar.

A key feature of our study is that we can investigate how workers respond to the possibility and experience of being punished. The possibility of punishment does not impact first-round worker effort. Workers exert approximately the same effort, on average, regardless of whether their manager can sanction them. This holds across worker race and gender. However, workers respond strongly to actually receiving punishment. Leveraging random-assignment, we causally identify that receiving a Round 1 sanction significantly reduces Round 2 effort. This discouragement effect increases with sanction size, making punishment counterproductive in this environment. These responses vary meaningfully by worker race and gender. White men are substantially more responsive to their manager's sanction decision than women or Black men. They increase effort more in response to their manager sparing them from possible punishment and decrease effort more steeply after being punished.

These repeated rounds mimic an environment where the worker must remain in the same workplace and under the same manager. However, in many settings, workers have the opportunity to exit the relationship. To shed light on this decision, we surprise the worker with the opportunity to opt into a third round with the same manager. Continuing guarantees weakly positive additional earnings, at a modest additional time cost. Workers who decline the third round are also asked whether they would have been (hypothetically) willing to play with a different manager, allowing us to distinguish relationship-specific exit from a general decline in willingness to work. We show that the larger the first-round punishment, the less likely workers are to continue the relationship with the same manager. However, the sanction has little effect on the willingness to participate with a new manager.

The effects are strongest for women.

Taken together, we find that when workers are effectively "stuck" in the relationship and must continue working for the same manager, women are less likely than men to reduce their effort in response to harsher punishment. But when given the opportunity to exit, women are more likely than men to leave the relationship following punishment. Punishment therefore produces different response margins across workers: men, particularly white men, respond more strongly through subsequent effort, even though this lower future effort may itself be punished. Women, on the other hand, do not adjust their effort in response to being punished, but are more likely to make a quiet but costly exit before Round 3 (Dana et al., 2006).

In sum, our evidence suggests that punishment fails to increase productivity and may lead to racial and gender disparities. In our setting, allowing for punishment does not increase effort, and when it is implemented, it can backfire, since white men reduce their effort. Women, on the other hand, exit relationships more often after being punished, even when these working relationships are weakly profitable. Simultaneously, punishment creates a racial disparity in the treatment of low-effort workers, since Black workers are sanctioned more harshly than equally unproductive white workers. At the group level, punishment increases the share of zero-surplus outcomes and reduces average surplus, with the losses falling on the worker, since their payoffs decline at no benefit for the manager. While monitoring alone increases worker effort, combining it with punishment destroys more value than it creates.

Our paper contributes to several strands of literature. First, we add to research on discrimination in workplace evaluations and punishment. While field studies show women and racial minorities sometimes face harsher consequences (Egan et al., 2019; Sarsons, 2017; Carvalho, 2025; Ferrario and Carvalho, 2026), our controlled design demonstrates that differential punishment can emerge even among workers with the same low effort, and even when managers' stated expectations do not differ across groups. We highlight that, at least

in our setting, these gaps are not driven by belief-based discrimination (Bordalo et al., 2016, 2019; Coffman et al., 2021, 2025), but instead suggest differential treatment of the same outcome. We also highlight the market-level consequences of differential punishment, showing that punishment matters for worker behavior, and that these effects vary by worker race and gender.

Second, we contribute to a broader line of research on punishment, incentives, and reciprocity. While prior work shows punishment can sustain cooperation (Fehr and Gächter, 1998, 2000; Carpenter, 2007), our results show it can also undermine cooperation: punished workers reduce future effort and are less likely to continue with the manager. This highlights important dynamic costs overlooked by focusing only on immediate effects.

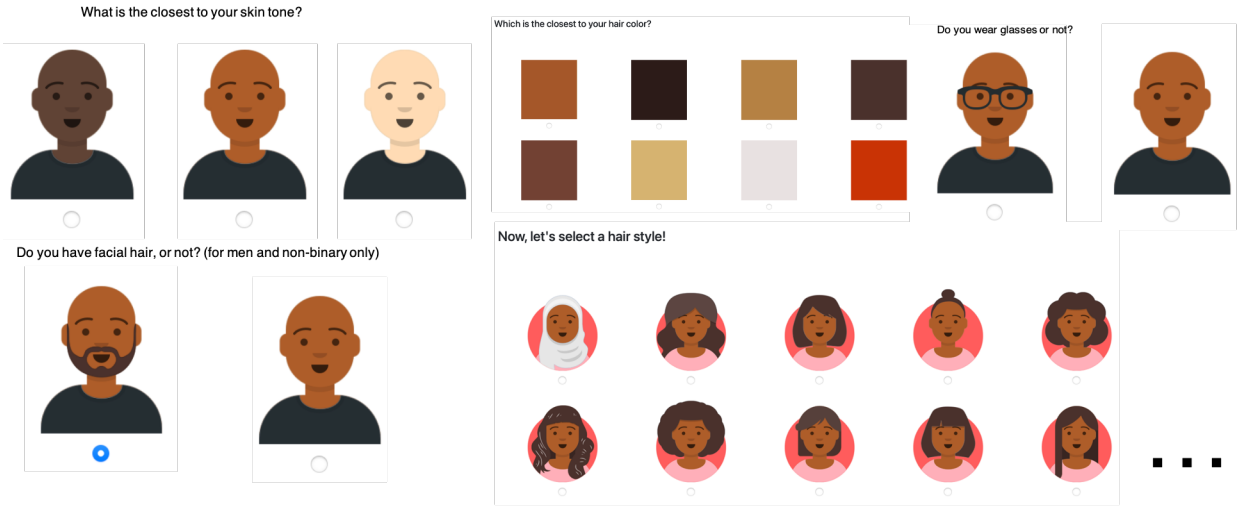
Finally, we contribute to the literature on organizational design and labor-market inequality. In our setting, managerial discretion to punish destroys value, increases the risk of poor outcomes, and generates unequal treatment of low-effort Black workers. The consequences vary: white men are more likely to retaliate through subsequent effort reduction, while women are more likely to exit. These findings suggest sanction-based institutions may lead to inefficiency, inequality, and costly exits.

2 Experimental Design

In our experiment, participants first complete an introductory survey and build an avatar that conveys their race and gender. They are then matched into manager-worker pairs and play a two-round effort game in which a good outcome benefits both members of the pair, and the manager may, in some treatments, punish the worker. We conduct the experiment online with real-time synchronous interaction between the two members of each pair. We targeted 3,200 participants, of whom a quarter (800) are Black, and the remainder are white.

Introductory Survey and Avatar Constructions Participants begin with a short demographic survey covering race, gender, education, and employment status, followed by an

Figure 1: Avatar choice



unincentivized elicitation of economic preferences: time and risk preferences, positive and negative reciprocity, altruism, and trust (Falk et al., 2018).

Because our design requires managers to observe worker race and gender, we need to convey these characteristics unambiguously while limiting the experimenter demand effects that heightened salience of race and gender might otherwise induce. We do so through an avatar, following Ruebeck (2025). At this stage, participants know only that they will play a game with another participant. Each participant builds an avatar that represents them throughout the study: the participant's answer to the gender question fixes the avatar's gender, and participants then choose the avatar's skin tone, hair color and style, facial hair (if male or non-binary), and whether to wear glasses. Figure 2 shows the sequence of choices. Embedding race and gender among several other choices mitigates their salience.

For the purposes of our analysis and in line with our pre-registration, we label participants who select the pale skin tone 'white' and those who select the medium or dark tones 'Black.' Reassuringly, fewer than 4% of participants choose a skin tone that appears inconsistent with the race they reported in the survey (see Table A1).

Slider Task and Effort Cost Elicitation Participants are next introduced to the slider task, the real-effort task that the worker will have to complete later. A slider is completed by dragging an indicator from the midpoint of a 0 to 100 scale to a randomly generated target. All participants must read the instructions and complete two sliders before proceeding.

We then measure each participant's cost of effort using a multiple price list. Participants choose, at each of a range of wages from \$0.05 to \$0.50 in \$0.05 increments, whether to complete a fixed number of sliders, S, for that wage or to complete none for nothing. For instance, a given row of the price list would ask a participant to choose whether to (1) complete S sliders and receive 0.30, or (2) do 0 sliders and receive nothing. Across three price lists, we set the fixed number of sliders, S, to 15, 30, or 45. Each participant is randomly assigned to see one of the possible price lists, and one row is selected at random and implemented at the end of the study, thereby incentivizing choices. Participants who chose to complete S sliders in the selected row must complete S sliders to receive the given wage and finish the study.

The Manager-Worker Game Participants are randomly assigned to roles, pairs, and treatment arms. Each participant is either a manager or a worker. All managers are White, while workers can be either Black or White. As a result, half of the groups are mixed-race (a Black worker and a White manager), and half are same-race (a White worker and a White manager).[3] Because the experiment requires two participants to play simultaneously, pairs are formed in real time from the pool of participants available at each moment, with the matching algorithm designed to maintain a 1:1 ratio of mixed-race and same-race pairs.[4] Pairs are assigned to one of four treatment arms, each containing a balanced number of mixed-race and same-race pairs.

Both members of the pair then learn their role, read the instructions, and learn that the

[3]While it would of course be interesting to study groups with Black managers as well, we made this design decision in order to maximize statistical power for studying differences across *worker* race.

[4]As participants arrive, each white participant is matched as a manager to a waiting Black worker when one is available, and is otherwise placed in a same-race pair. Details of the matching procedure are provided in Appendix A.1.

game consists of two rounds, one of which is randomly chosen to determine payoffs. They answer comprehension questions and cannot proceed until they answer correctly. Before the worker begins, the manager completes a short "kick-off" task of counting zeros in a matrix; this task does not affect either player's payoff but is intended to increase the manager's engagement and sense of ownership in the game.

The worker then has 180 seconds to complete up to 50 sliders. Each completed slider raises the probability of a good outcome by one percentage point from a baseline of 5%, so a worker who completes all 50 sliders raises this probability to 55%. A good outcome pays both players 10 tokens, and a bad outcome pays the worker 10 tokens and the manager 0. This creates an incentive for the worker to "shirk," as their payoff in the good and bad state is the same. However, the worker may also have reason to exert effort; they may care about the manager's payoff (other-regarding preferences) or, if punishment is possible, they may try to avoid punishment. While the worker works, the manager reports how many sliders they expect the worker to complete and how many they think the worker should complete, allowing us to study the relationship between managers' expectations and their subsequent decisions.

Treatment Variations We implement a 2×2 design that varies whether the manager can punish the worker and whether the manager observes the worker's effort.

In the punishment arm, the manager can deduct up to 10 tokens from the worker at the end of the round, after the outcome is revealed. Punishment is not costly to the manager, but it is a social loss, as the manager does not keep the tokens deducted from the worker. Workers are informed at the start of the first round whether their manager can punish them. This variation lets us study how anticipated punishment affects worker effort and how past punishment affects future effort.

In order to maximize statistical power for detecting differential punishment according to worker race, we introduce the possibility of punishment to all *managers* at the start of the

game.[5] At the start of the first round, every manager is told that they can deduct up to 10 tokens from the worker, and that this sanction will be imposed with 50% probability. We ask all managers to make a punishment decision in the first round and only after revealing whether they are indeed in the punishment arm. Only decisions from those managers assigned to the punishment arm are implemented. Managers in the no-punishment arm do not have their round 1 decisions implemented, and they make no further punishment decisions.

The other treatment variation regards whether managers observe workers' effort. If the effort is revealed, the manager observes both the outcome and the number of sliders the worker completed at the end of each round. If effort is hidden, managers observe only whether the outcome is good or bad. Managers know from the outset whether they will observe effort. Workers are told at the start of the first round that the manager observes their effort with 50% probability, and that observation in the first round implies observation in the second. In the hidden-effort arms, we additionally ask the manager, after their punishment decision, to guess how many sliders the worker completed. The main goal of this variation is to test whether (biased) beliefs about effort drive punishment decisions, and how those compare with certainty on effort. Although testing classic moral hazard issues is not the study's primary goal (Holmstrom, 1982; Hoppe and Schmitz, 2018), this design also lets us examine how effort revelation affects effort provision in the no-punishment treatment arm.

Note that by the start of the second round, both managers and workers know their exact treatment arm. Finally, before the second-round outcome is revealed, we ask the worker whether they would like to play a third round with the same manager, identical to the first two and paid on top of the randomly-selected round. Workers who decline are asked whether they would instead have been willing to play with a different manager.[6] These choices allow us to study the effects of punishment and monitoring on workers' labor supply, both on the intensive margin (effort exerted in the second round) and the extensive margin (willingness

[5]Note that, in contrast, only workers assigned to the punishment arm ever learn about the possibility of punishment.

[6]Unfortunately, it was not feasible to incentivize this additional choice, as repairing managers and workers at this stage would have been impractical.

to participate in the third round).

Implementation We recruit participants through Prolific, restricting the sample to participants located in the United States, aged 18 or older, with at least 100 prior submissions and an approval rating of 95% or higher, who self-identify as white or Black/African-American. We use Prolific's screening to oversample Black participants so that the recruited sample is three-quarters white and one-quarter Black. This way, we target 3200 participants who complete the study, forming 1600 pairs, half mixed-race and half same-race, evenly split across the four treatment arms.

We pre-registered our study with the AEARCT registry.[7] Note that our pre-registration specifies race as our primary focus, with gender differences as a secondary focus.[8] To streamline writing and presentation, we present results by race and by gender simultaneously below.

Participants must pass several attention and comprehension checks distributed throughout the session. We pre-registered our exclusion criteria and implement them as follows. We excluded from the sample (i) any participant who has failed two attention checks[9],(ii)

[7] `https://www.socialscienceregistry.org/trials/17113`, trial 17113. Our first set of pre-specified research questions focus on managers. Do managers punish Black workers more than white workers for negative outcomes? And, if so, are these differences well-explained by managers holding different beliefs about worker effort? Our primary outcome measures for these questions are punishment amounts (# of tokens) levied during Round 1 across treatment and worker race, conditional on project outcome. To test the role of beliefs, we look directly at Round 1 manager expectations and also compare punishment decisions across the Hidden-Effort and Known-Effort arms. We pre-specified using Round 2 beliefs and punishment decisions to further investigate the motivations for punishment. Our second set of pre-specified research questions ask how the anticipation of punishment shapes the labor supply decisions of workers? Do they anticipate harsher punishment, and if so, does this impact their decisions about how much effort to invest in their work or how willing they are to continue working under that manager? To test the effects of anticipated punishment, we compare effort exerted in Round 1 across treatment and worker race. In addition, we look at the follow-on effects of punishment by investigating effort in Round 2 and Round 3 opt-in decisions, by treatment, race, and Round 1 punishment decisions. Finally, we explore the combined impacts of these channels on total group earnings by treatment and race. These pre-specified questions and outcomes guide our analysis below.

[8] We hypothesized that the gender gap could plausibly go either way; male workers could receive greater punishment than female workers (consistent with a manager belief that women may work harder/be more conscientious than men), or male workers could receive less punishment than female workers (perhaps due to differences in perceived power, hierarchy, or status).

[9] Initially, the pre-registration stated that any participant who failed an attention check would be dropped. However, to comply with Prolific's guidance, the code dropped participants only after their second attention check failure. Upon realizing this inconsistency, we updated the pre-analysis plan to align with the implemented code and Prolific's guidelines. At the time of this change, we had already collected 35% of

any participant who dropped out during the study, and (iii) any participant whose partner dropped out during the study. Participants who complete the study receive a fixed participation fee of $4 plus bonus payments. First, in the multiple-price-list task, we randomly select one row, and participants are paid according to the accept/reject choice they made for that row. Second, they earn $0.10 per token obtained in the game: in each of the first two rounds, a worker earns up to 10 tokens (reduced by any sanction imposed by the manager), and a manager earns 10 tokens following a good outcome and 0 otherwise, with one of these two rounds selected at random for payment. Finally, if the worker chooses to play a third round, both participants are additionally paid $0.10 per token earned in that round.

3 Data and Summary Statistics

Sample construction Table A2 shows the sample from recruitment to the dataset used in the analysis. Of the 6334 participants who began the study, 77% were matched into manager-worker pairs, and 50% completed all rounds without dropping out or having their partner drop out. Of the individuals who started the study, 1461 participants (23%) arrived but were never matched to a partner; among matched participants, 909 (14%) dropped out themselves, and 816 (13%) were lost because their partner dropped out. Attrition is similar across roles (34% for managers and 33% for workers).

Sample composition Table A3 reports baseline characteristics by role. Throughout, worker race is defined by avatar skin tone. The avatar closely tracks a worker's own reported race: 90% of Black-avatar workers and 99% of White-avatar workers self-report the corresponding race/ethnicity. The means of managers and workers differ on several observables (not formally tested), but this is completely explained by the fact that Black and white individuals are different, and all managers are white, and workers can be both. The

the data. We applied the revised rule retroactively to ensure all participants in the sample meet the new criteria. Restricting the sample to participants who failed no attention check, as originally pre-registered (948 workers; 488 managers in the Known-Effort arm), leaves our main results unchanged (Tables A41–A44).

difference in comprehension mistakes is mechanical: managers make more comprehension mistakes throughout the study than workers. In Table A4, we can also observe that black and white workers differ in many observables. Black workers are 6 years younger, less likely to have a university degree, and more likely to be employed. They also report slightly higher risk tolerance, altruism, and negative reciprocity, and lower trust. They also have a lower measured effort cost (0.88 vs. 0.97 cents per slider). Similarly, male and female workers also differ. Males are more likely to be employed and make more comprehension mistakes than females. Male workers exhibit more risk tolerance, patience, and negative reciprocity. However, they report lower positive reciprocity and altruism.

Balance across treatment arms. Table A5 reports means of the observed characteristics by treatment arm. Randomization is successful, since no covariate differs significantly across arms at the 5% level, and the share of participants assigned to a mixed-race pair is close to one-half in every arm, confirming that the real-time matching algorithm preserved the intended mixed/same-race ratio without inducing imbalance. Tables A6 and A7 show that balance also holds within role, for managers and workers separately. A handful of point estimates are individually significant, but their number and size are consistent with what would arise by chance under this many comparisons.

Attrition is different across treatment arms (Table A8). In the arm No-Punishment×Known-Effort, almost 30% of participants did not complete the study. In the other treatment arms, incompletion rates are similar, between 34% and 36%. This cross-arm variation is significant at the 10% level ($p = 0.055$) but, reassuringly, symmetric across roles. Partner-driven dropout, which is the exclusion that is not mechanical at the pair level, is balanced across arms ($p = 0.12$). Attrition is, however, mildly selected on race. Black participants are more likely to drop out compared to white ones. However, dropped and retained participants are indistinguishable on age, gender, education, and employment (Table A9).

4 Results

Our experiment allows us to study both sides of the market: how managers decide to punish and how workers respond to both the possibility and realization of punishment. First, we discuss manager decisions; then, we turn to worker decisions. Finally, we discuss the implications of these findings for market efficiency and profit distribution.

4.1 Manager Decisions: The Supply of Punishment

In this section, we study what drives managers' decisions to punish and whether punishment varies by workers' race and gender. First, we document the overall supply of punishment: how often and how heavily managers sanction, and how punishment responds to the round's outcome and observing workers' effort. Then, we analyze whether managers punish Black and white workers and male and female workers differently, conditional on what they observe. Finally, we study the beliefs and reasoning behind managers' decisions, particularly how managers' expectations about effort and the justifications they offer for their choices can account for the documented findings.

4.1.1 Determinants of Punishment

We begin by asking how much punishment managers levy, and what drives their decisions. Recall that in Round 1, every manager makes a punishment decision, knowing that it will be implemented with 50% probability. This design choice lets us collect punishment decisions from all managers, not just those in the Punishment arm, increasing the analysis's statistical power.

Overall, punishment is rare and small in scale. Only 28% of managers impose any sanction at all, and the average sanction is just 1.51 of the 10 tokens available (Table A10). Conditional on punishing, the average sanction is 5.46 tokens. These low levels of punishment are perhaps surprising, given that punishment is free and bad outcomes are very

common (68% of managers face a bad outcome in the first round, as presented in Table 1). Even conditional on a bad outcome, only 37% of managers choose to sanction their worker, with an unconditional average deduction of 2.07 tokens. When they punish, they do so by 5.64 tokens. One possibility is that managers refrain from punishment in part because of fear that workers will reciprocate with lower effort in Round 2, in line with the constraining impacts of counter-punishment documented in (Nikiforakis, 2008). Surprisingly, 9% of managers who receive a *good* outcome nonetheless punish their worker. This is more common when effort is observed (12% in the Known Effort arm) but still sometimes occurs when it is hidden (5%). This is the first indication that information about effort helps explain managers' behavior.

Table 1: The Choice Environment: Outcomes and Punishment

Round-1 outcome	N	% of rounds	% punishing	Mean sanction	Mean | punished
Bad outcome	1062	67.5	36.7	2.07	5.64
Good outcome	512	32.5	8.6	0.33	3.86
All	1574	100.0	27.6	1.51	5.46

Note: This table describes the Round-1 environment managers face, one observation per manager. Outcomes are classified as Bad or Good depending on the realization of the joint project in Round 1. N is the number of managers, % of rounds is the share of managers facing each outcome, % punishing is the share who choose a strictly positive sanction, and Mean sanction is the average number of tokens deducted. Mean | punished is the average sanction among managers who chose a strictly positive sanction.

Indeed, observing effort is a key determinant of sanctions. Managers randomly allocated to Known-Effort arms are 8 pp ($p < 0.01$) more likely to punish and sanction almost 0.4 more tokens than their Hidden-Effort counterparts. Note that this is not driven by differences in worker effort across treatments. Conditional on both worker effort and observed outcome, managers who had access to information on worker effort punished more (Table 2). Naturally,

there is no difference across punishment arms, since managers only learn whether their sanction is implemented after making their decision, and ex-ante there is a 50% probability that it will be.

Table 2: Punishment Decisions by Treatment Arm

	Punished	Sanction	Punished	Sanction	Punished	Sanction
Punishment arm	-0.012	0.045	-0.012	0.043	-0.009	0.059
	(0.022)	(0.147)	(0.022)	(0.146)	(0.021)	(0.141)
Known effort arm	0.080***	0.387***	0.080***	0.384***	0.068***	0.311**
	(0.022)	(0.147)	(0.022)	(0.145)	(0.022)	(0.141)
Worker's effort			-0.005***	-0.034***	-0.003***	-0.018***
			(0.001)	(0.007)	(0.001)	(0.007)
Good outcome					-0.261***	-1.621***
					(0.020)	(0.120)
Constant	0.241***	1.287***	0.380***	2.162***	0.405***	2.314***
	(0.019)	(0.118)	(0.032)	(0.219)	(0.032)	(0.215)
R^2	0.008	0.004	0.027	0.022	0.098	0.086
Observations	1,574	1,574	1,574	1,574	1,574	1,574

Note: This table shows the effect of each treatment on punishing decisions. Punished is a binary variable that equals one if the manager applied any sanction. Sanction represents the number of tokens the manager fined the worker. Columns 3–4 add the worker's Round-1 effort (sliders completed); columns 5–6 additionally add the Round-1 outcome. Heteroskedasticity-robust standard errors in parentheses. Asterisks denote pointwise statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.1.2 Gender and Racial Punishment Gaps

We now turn to one of the study's central questions: do managers punish workers differently by race and gender? We start by analyzing the unconditional gaps, and then account for

the round characteristics.

Unconditionally, managers punish Black workers more. Table A11 shows that managers are 4.4pp ($p < 0.05$) more likely to sanction a Black worker and deduct 0.3 more tokens on average ($p < 0.05$). This gap equals roughly 20% of the mean sanction. We find no meaningful or statistically significant difference by worker gender.[10]

A natural concern is that this raw gap reflects differences in outcomes, rather than the worker's race. Indeed, conditional on the outcome of the game, we find no racial gap in punishment (see columns (1) and (2) of Table 3). We also continue to find no evidence of a gender gap. We find that the final outcome drives punishment decisions, with managers punishing significantly less by about 1.7 tokens ($p < 0.01$) when the outcome is good. Observing effort increases punishment by an average of 0.3 tokens ($p < 0.05$).

[10]This contrasts with field evidence documenting harsher punishment for women.

Table 3: Punishment Decisions: Treatment and Worker's Characteristics

	S_{g1}				
	(1)	(2)	(3)	(4)	(5)
Black worker	0.304**	0.184	0.177	-0.083	-0.243
	(0.147)	(0.202)	(0.202)	(0.244)	(0.241)
Male worker	0.035	0.132	0.143	0.134	0.119
	(0.150)	(0.208)	(0.208)	(0.250)	(0.243)
Good outcome (G_{g1})		-1.688***	-1.676***	-1.689***	-1.726***
		(0.177)	(0.178)	(0.177)	(0.179)
Black × Good outcome		0.041	0.047	0.072	0.041
		(0.234)	(0.234)	(0.234)	(0.233)
Male × Good outcome		-0.143	-0.144	-0.143	-0.131
		(0.237)	(0.237)	(0.237)	(0.235)
Known effort arm			0.307**	0.045	0.053
			(0.141)	(0.223)	(0.219)
Punishment arm			0.067	0.069	0.088
			(0.142)	(0.142)	(0.138)
Black × KE arm				0.497*	0.515*
				(0.283)	(0.278)
Male × KE arm				0.033	0.031
				(0.289)	(0.282)
Constant	1.339***	1.922***	1.728***	1.859***	2.093**
	(0.117)	(0.163)	(0.189)	(0.203)	(0.938)
Manager controls	No	No	No	No	Yes
Observations	1,574	1,574	1,574	1,574	1,574

Note: OLS of the Round-1 sanction S_{g1} (tokens deducted, 0–10) on worker race and gender and their interactions, one observation per manager, all arms pooled. *Black worker* and *Male worker* denote the worker's avatar race and gender (base: White, female). Column 1 reports the unconditional gaps; column 2 adds the Round-1 outcome and its interactions (*Black/Male* × *Good outcome*; base: bad outcome), so that *Black worker* becomes the gap after a good outcome and a negative interaction indicates a larger gap after bad outcomes; column 3 adds the treatment-arm indicators; column 4 adds their interactions with the Known-Effort arm (base: Hidden Effort); column 5 adds manager controls (demographics, elicited preferences, the MPL effort cost and list fixed effects, and attention/comprehension counts). Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

However, we also observe that the monitoring effect is concentrated entirely on Black workers. Once we allow the treatment effect to vary by race, the Known-Effort effect is essentially zero for white workers; only the Black $\times$ Known-Effort interaction is significant ($p < 0.10$). When managers observe effort, they punish Black workers 0.5 tokens more than white workers, with no analogous gender gap. Interestingly, information *amplifies* rather than disciplines the racial gap.

In Table 4, we explore this pattern further by focusing specifically on the Known-Effort treatment and looking at heterogeneity by worker effort. Overall, we find that one extra completed slider decreases punishment by 0.05 tokens ($p < 0.01$, column (2)). More interestingly, the racial gap is most pronounced at lower effort levels. At low effort (0 sliders), Black workers are punished 1.3 tokens more than white workers ($p < 0.05$, column (4)). Managers cut Black workers' sanctions faster as effort rises ($-(-0.016 - 0.049) = 0.065, \quad p < 0.05$), but only because they start from a higher sanction at low effort. The gap closes only once a Black worker completes about 27 sliders, just above the average effort level of 26. Equally low-performing individuals face different sanctions by race. We find no parallel pattern by gender.

Table 4: Punishment Decisions: the Role of Effort

	S_{g1}			
	(1)	(2)	(3)	(4)
Black worker	0.438	0.240	1.467**	1.303**
	(0.286)	(0.281)	(0.607)	(0.585)
Male worker	0.135	0.137	0.354	0.363
	(0.297)	(0.291)	(0.609)	(0.586)
Good outcome (G_{g1})	-1.745***	-1.485***	-1.652***	-1.775***
	(0.252)	(0.254)	(0.259)	(0.269)
Black × Good outcome	-0.020	-0.080	0.177	0.177
	(0.339)	(0.338)	(0.335)	(0.343)
Male × Good outcome	-0.056	-0.061	0.014	-0.000
	(0.347)	(0.346)	(0.343)	(0.345)
Effort (N_{g1})		-0.046***	-0.018	-0.016
		(0.010)	(0.017)	(0.017)
Black × N_{g1}			-0.051**	-0.049**
			(0.020)	(0.020)
Male × N_{g1}			-0.009	-0.010
			(0.020)	(0.020)
Constant	1.956***	3.162***	2.440***	3.149**
	(0.237)	(0.367)	(0.526)	(1.448)
Manager controls	No	No	No	Yes
Observations	799	799	799	799

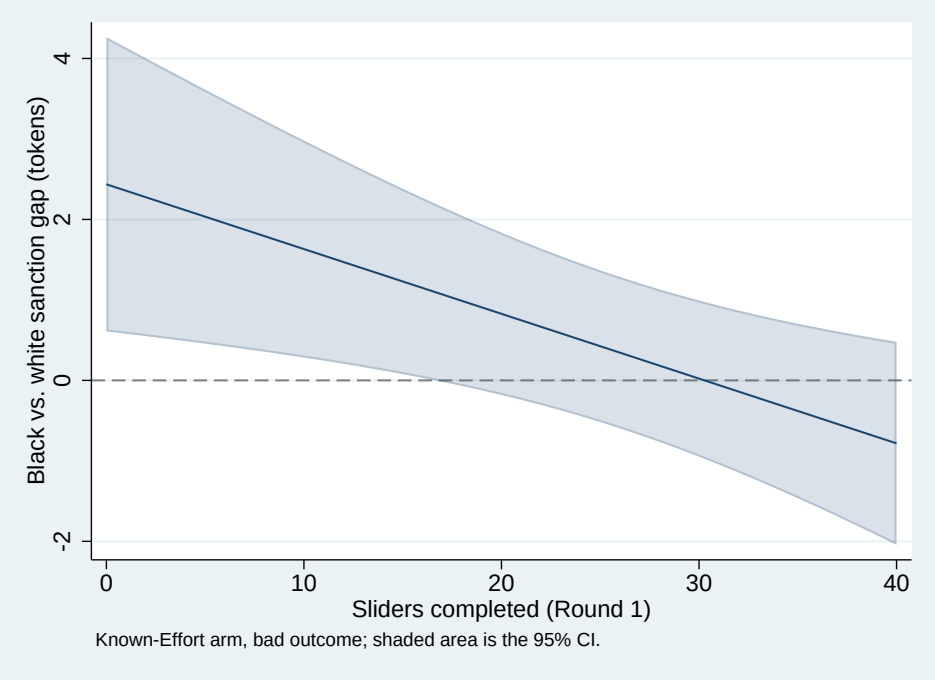
Note: OLS of the Round-1 sanction S_{g1} (tokens, 0–10), restricted to the Known-Effort arm, where the manager observes the worker's actual effort N_{g1} before choosing the sanction; one observation per manager. Column 1 conditions on the outcome and its race and gender interactions; column 2 adds observed effort N_{g1} (sliders completed); column 3 adds *Black/Male* × N_{g1}, letting the effort slope of the sanction differ by worker race and gender; column 4 adds manager controls (demographics, elicited preferences, the MPL effort cost and list fixed effects, and attention/comprehension counts). *Black worker* and *Male worker* denote the worker's avatar race and gender (base: White, female); a residual *Black worker* coefficient, after conditioning on observed effort, is not accounted for by lower observed effort. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We also observe that the racial gap is on the extensive margin - whether to punish or not - rather than on the severity of the sanction (Table A13). At low effort (0 sliders), managers are 16.5pp ($p < 0.05$) more likely to punish Black workers, but conditional on punishing, we find no statistically significant differences in sanction levels by race (a large but imprecise 1.31 tokens). Additionally, evidence suggests managers are more likely to sanction the maximum possible (10 tokens) if the worker is Black. At low effort (0 sliders), Black workers are 8pp ($p \approx 0.15$) more likely to receive 10-token sanctions. Although underpowered, the estimate should be compared to a 7.6% base rate. This corresponds to doubling the likelihood of receiving the maximum punishment.

Splitting the sample further reveals that the racial gap is most pronounced among lower-effort *men* (Table A12). We look separately at men, women, Black, and white workers and, conditional on the outcome, analyze how effort interacts with gender or race within each population. Among men, Black workers are fined 2.4 tokens more than White men at zero effort ($p < 0.01$). The gap closes as effort rises, by 0.08 tokens per slider completed ($p < 0.01$). For this population, assuming linearity, the punishment gap closes only when a worker completes about 30 sliders, 4 sliders above the average of 26 (see Figure 2).[11] Among women, by contrast, the Black–white gap is smaller and not statistically significant.

[11]Note that the gap becomes statistically indistinguishable from zero beyond roughly 17 sliders, at 5% and 19 at 10%.

Figure 2: Black-white punishment gap among men, by observed effort



Note: Marginal effect of a Black (vs. white) male worker on the Round-1 sanction, from the Known-Effort specification of Table A12, evaluated at a bad outcome across levels of observed effort. The shaded area is the 95% confidence interval. The gap is statistically significant (the band excludes zero) only below about 14 completed sliders; the point estimate reaches zero at roughly 31 sliders, above the sample mean of about 26.

Managers' sanctions also respond to effort differently across groups. For Black workers, an additional completed slider significantly reduces sanctions by 0.05 tokens for Black women ($p < 0.01$) and by 0.09 tokens for Black men ($p < 0.01$). White workers' sanctions do not move with effort at all. This racial difference in effort sensitivity is itself significant among men (-0.08, $p < 0.02$).

In summary, the racial punishment gap operates through the treatment of lower-effort men. Managers appear most lenient with lower-effort white men and most punitive with lower-effort Black men. The gap does not depend on the manager's own gender: female and male managers both sanction lower-effort Black men more than white men, and neither the gap nor its closing with effort differs significantly between them (Appendix Table A14, Panels A and B).

4.1.3 Beliefs, expectations, and stated reasoning

With these punishment patterns in mind, we now ask how managers' beliefs about effort explain their punishment decisions and how they justify them. We also study whether these factors help account for the racial gap.

While the worker completes the task in each round, we ask managers: (i) how many sliders they believe the worker *should* do, and (ii) how many they think the worker *will* do. On average, managers believe workers should complete 33.5 out of 50 sliders. However, managers also expect them to fall short and believe workers will do just under 28 (Tables A15 and A16). In the Hidden-Effort treatment arms, we also ask managers to estimate how many sliders the worker completed after observing a good or bad outcome and after making their punishment decisions. After seeing a bad outcome, managers estimate the worker did 18 sliders on average; after a good outcome, they estimate 29. Overall, workers' race or gender does not significantly predict these expectations.[12]

Tables A17 and A18 show that these beliefs matter for punishment decisions. In the Hidden-effort arms, the manager never learns the true effort, so decisions can depend only on expectations and beliefs. Instead, in the Known-effort arms, managers learn how many sliders the worker completed. Thus, their decisions can account for the difference between expectations and reality. We observe that the difference between how many sliders a worker *should* do and either the manager's belief (0.034 tokens, $p < 0.01$) or the worker's actual effort (0.019 tokens, $p < 0.01$) increases the levied sanction.[13]

Importantly, these expectations seem equally informative for sanctions against Black and white workers in the Known-Effort arms. There is no significant interaction between race and expectations. In the Hidden-Effort arms, on the other hand, there is suggestive evidence

[12]The exceptions are that managers believe that males *should* do more sliders than female workers, but they do not think they *will*; and in the case of a good outcome, managers also believe Black workers have completed 3.69 more sliders than white workers.

[13]Interestingly, the difference between their guess about how many sliders the worker completed before and after learning the outcome (HE arm) also predicts the sanction amount (0.022, $p < 0.05$), but this loses importance once the manager learns the true effort (0.004, not statistically significant).

that the difference between what a manager believes a worker should versus will do matters more for punishment decisions of Black workers compared to white workers. These point estimates are noisy, but their size is meaningful (0.025 for $Black \times (Should - \hat{N}_{g1})$, $p < 0.14$, and 0.031 for $Black \times (Will - \hat{N}_{g1})$, $p < 0.10$).

A different reading of the gap is that managers believe punishment works better on Black workers, so that a higher sanction is a justified choice rather than a sign of bias. This story predicts a gap in Round 1 but not in Round 2, when a sanction can no longer affect future effort, which is what we observe, so the timing alone cannot rule it out. Two pieces of evidence can. First, if managers sanctioned Black workers because they expected a larger effort response, their stated expectation of the worker's Round-2 effort should rise more after sanctioning a Black worker. We find that it does not. In the Known-Effort arm with male workers, the interaction between the sanction and worker race is zero (Table A19), and in the full sample it is negative: managers who punish a Black worker lower their expectation of his effort by about eight sliders more than managers who punish a white worker, with no such difference in the placebo arm where the sanction is chosen but not applied. Second, the gap is not carried by the managers who say they punish to motivate. Among the 10 percent of managers whose stated reason is motivation, Black and white workers are sanctioned alike (Table A23).

An important caveat of the Hidden Effort scenario is that managers were asked to guess the number of sliders completed after deciding on punishment. We did not ask them beforehand because we did not want to prime their decisions. We cannot distinguish between a manager who formed an expectation of the number of sliders completed before deciding on punishment and a manager who post-rationalized their punishment decisions with their stated beliefs.

Managers' own explanations of their sanctions improve our understanding of the rationale for the observed gap. Using a language model to sort the free-text explanations into categories, we find that the gap against low-effort Black men is concentrated among managers

who describe their decision as an evaluation of the worker's performance, past or expected. Managers who invoke fairness, empathy, or their own earnings sanction Black and white workers alike. The text of the explanations points the same way: a Black man's low effort is more often attributed to his character, a white man's is more often excused or treated as something to encourage. Appendix A.2 reports the classification and these results in detail.

4.1.4 Insights from Round 2

The previous analysis relied only on Round 1 decisions, where endogeneity is least threatening. However, Round 2 might help clarify the dynamic path of punishment and shed light on the underlying mechanisms. In Round 2, managers have more information about the paired worker, including full knowledge of whether they can use punishment as a tool and whether the worker knows their effort is observed. In the Known-Effort arm, they also know how the worker changed their effort behavior between rounds. We use this information to study whether managers maintain the racial gap for low-effort workers. Since only managers in the Punishment treatment arms decide on sanctions in Round 2, our analysis focuses on them. In summary, we will restrict this analysis to managers in the Known-Effort arm with Punishment.

Overall, in this treatment arm, punishment rates fall (from 32% to 27%), and the average sanction remains constant (from 1.7 tokens to 1.68 tokens). This means that the managers who punish are punishing more severely (5.3 to 6.3 tokens). We are currently interested in whether managers continue to disproportionately penalize Black workers. Table A26 shows that they do not. If, in Round 1, a zero-effort Black worker received a sanction 1.63 tokens higher, we do not find this difference in Round 2. The same holds for the low-effort Black men who drove the first-round result. Their zero-effort penalty moves from roughly +2.3 to −1.2 tokens.

Interestingly, the manager begins to treat effort as the primary driver of the sanction. In Round 1, effort did not predict sanctions, but in Round 2, an increase in effort by one

slider leads to a decrease of 0.125 tokens in the sanction ($p < 0.01$), and the raw correlation between the sanction and observed effort roughly doubles, from -0.24 to -0.50. The second-round sanction tracks effort itself, and we no longer find a role for race. Consistent with this, managers also reward improvement, regardless of race (Table A27). Regressing the change in the sanction between rounds on the change in the worker's effort, we observe that each additional slider completed relative to Round 1 lowers the sanction by 0.08 tokens ($p < 0.01$). Since the low-effort workers who improve the most between rounds are disproportionately Black, crediting effort alone helps explain why the first-round gap closes.

In summary, the second punishment decision is better informed and differs from the first regarding the low-effort racial penalty. We no longer observe disproportionate punishment of Black workers, effort is priced directly, and improvement is rewarded independent of race.

In the next section, we discuss why we detect the racial gap only in the first round, and what that implies about the nature of the bias.

4.1.5 Discussion

The supply-side results leave us with a pattern to explain. Racial differences in punishment arise when the manager *observes effort* but not when effort is hidden. They also concentrate at *low effort*, where a Black worker near the bottom of the effort distribution is sanctioned by a token more than an otherwise-identical White worker, and Black men are punished by two tokens more. This gap is mostly explained by willingness to punish rather than by punishment intensity. We do not find the same effect in the second round of the interaction. In this section, we discuss which mechanisms might explain this behavior.

The first candidate, for which there is an extensive literature, is statistical discrimination. If Black workers are, on average, less productive at the task, a manager who cannot perfectly observe effort might treat race as a proxy for productivity. The data provide evidence against this channel. The gap appears exactly when effort *is* observed and vanishes when effort is hidden, which is when race could serve as a proxy. The punishment gap is documented

conditioning on observed effort, so equally low-performing Black and white workers are punished differently. It is not that Black workers do less and are sanctioned for it.

Then, we studied the role of expectations and beliefs. Managers do not hold different expectations for Black and white workers in the Known-Effort arm, and those who punish a Black worker do not expect a larger effort response from him. If anything, they expect less. The explanations managers give for their sanctions locate the gap more precisely. Across all managers, the stated reasons are distributed similarly by worker race. But the gap against low-effort Black men is concentrated among managers who describe their decision as an evaluation of the worker's performance, while managers who reason from fairness, empathy, or their own payoff treat Black and white workers alike. The wording differs in the same direction: a Black man's low effort is more often attributed to his character, a white man's more often excused or framed as something to encourage, although these differences in language are suggestive rather than precisely estimated.

Finally, we need to consider taste-based discrimination. If managers simply disliked Black workers, independent of performance, we would likely observe this pattern across treatments and specifications. However, the punishment gap concentrates at low effort. This might suggest that managers need a pretext to punish Black workers more harshly. It might also explain why we do not observe the gap in the second round: there was an overall increase in the average number of sliders completed, as if the 'excuse' were gone. This is consistent with Dana et al. (2007), for example, who show that people choose to be fair because they dislike appearing unfair to others. However, as soon as there is wiggle room, they show less fairness concern.

A related possibility is in-group leniency rather than out-group animus: all managers in our experiment are white, so the gap could reflect forgiveness toward low-effort white men rather than harshness toward low-effort Black men. The sanction levels separate the two. Among managers who judge performance, a white man who completed no sliders is fined about two tokens, while a Black man is fined about six. Among managers who give

other reasons, both are fined close to zero. Judging performance therefore adds little to the sanction of a white man and a great deal to the sanction of a Black man. The gap is additional punishment of Black men, not a discount for white men. Manager traits point the same way. Managers with a stronger taste for retaliation sanction Black workers more, but not white workers, and this holds when the stated-reason channel is in the same regression (Table A28). Altruism is associated with a larger gap in the full sample (Table A29) but not among the male workers, where the gap is concentrated.

Finally, the evidence that the effect is mostly concentrated on low-effort Black men may reflect negative stereotypes around work ethic and character (Boskin, 1988; Collins, 2004; Lemons, 1977). The same low slider count may be read as diagnostic of the worker's type when he is a Black man, but discounted as noise or bad luck when he is a White man. Since the manager's expectations do not differ by race, what may differ is the interpretation or meaning attributed to the effort signal. This makes our setting different from Coffman et al. (2021), where discrimination tracks stated beliefs about average group performance. Another feature of stereotypes is that individuals rely more on them when information is scarce (Bordalo et al., 2019; Coffman et al., 2024). This is somewhat consistent with the dynamics we observe in the experiment. We find the punishment gap only in Round 1, when the manager observed only the workers' avatars and effort. In Round 2, the manager observes the worker's improvement and starts tracking and rewarding effort itself, independently of the worker's race.

4.2 Worker Decisions: Anticipation of and Responses to Punishment

In this section, we study how workers adjust their labor supply in response to the possibility and realization of punishment. We first examine how anticipating possible punishment shapes initial effort choices in Round 1. Then, we study how workers' Round 2 effort responds to realized punishment decisions. Finally, we study the extensive-margin decision of

whether to continue working for the same manager in an optional Round 3, asking whether earlier punishment increases the likelihood of opt-out. Throughout, we focus on two dimensions of heterogeneity that are salient in the workplace and in the discrimination literature: race and gender.

4.2.1 Anticipation of punishment and initial effort

We first ask whether the possibility of punishment changes initial effort provision among workers. We compare Round 1 effort decisions (number of sliders completed) across our two randomly-assigned treatments: No Punishment and Punishment. In the Punishment treatment, workers know their manager can sanction them after observing the Round 1 outcome. In the No Punishment treatment, the worker instructions do not mention the possibility of punishment. All other aspects of the treatments are held constant, including the fact that all workers are uncertain whether their manager will observe the exact number of sliders they complete in Round 1. A large experimental literature on punishment in public goods games and other social dilemmas suggests that the possibility of punishment likely increases effort (Fehr and Gächter, 2000; Carpenter, 2007).

And yet, we observe no effect of the possibility of punishment on initial effort decisions. On average, workers complete 26 sliders in Round 1 (SD of ≈ 11), just over half the total amount possible. This is nearly identical across treatments. The mean number of sliders completed varies by less than 0.5 sliders across treatments ($p > 0.9$).[14] Given the baseline probability of a good outcome (5%) and the returns to effort (1pp per slider), this translates to an expected likelihood of a good outcome of just under one-third in the first round.

Why might effort decisions fail to respond to anticipated punishment in our setting? A few possibilities seem worth considering. First, workers may have been aware of the possibility of punishment but did not expect managers to actually punish them. This is

[14]The distributions are also indistinguishable across treatments, see Appendix Figure A.3. And the share of workers who do no sliders (approximately 1%) and the share of workers who do all 50 sliders (approximately 1.5%) are also unchanged across treatments.

somewhat internally consistent with the overall low rates of punishment we indeed observe in Section 4.1. However, we cannot test this explanation directly. We did not ask workers about their expectations of punishment, as we felt that direct questions on anticipated punishment could raise concerns about experimenter demand or priming effects. A second possibility is that psychological forces offset the financial incentive to increase effort that possible punishment provides. For instance, the possibility of punishment might shift how workers feel about their manager, reducing altruism or goodwill toward the other party; it might also shift how they feel about the work context itself, creating discouragement or low morale (Bowles and Polania-Reyes, 2012). Again, we do not have data that allows us to test these explanations directly.

More broadly, we can ask what factors do seem to matter for Round 1 effort decisions. In Table A30, we predict Round 1 effort from a battery of information we have on workers, including their demographics (race, gender, age, education, employment status), economic preferences (as measured by our introductory survey), and other study behavior. We find many of these factors play a significant role. First, we see clear demographic differences, with Black workers and older workers completing significantly fewer sliders on average. Second, the impact of measured economic preferences is both sizable and intuitive. More risk-tolerant participants complete significantly fewer sliders, while participants who more strongly endorse positive reciprocity complete more sliders; both of these effects are stronger in the punishment treatment than in the no-punishment treatment. Negative reciprocity predicts less effort, but only in the no-punishment treatment, suggesting that the possibility of punishment may motivate exactly those workers who would expect to punish if in the manager's position. These patterns are consistent with the idea that effort in our setting has both financial and relational components: while some workers may be responding primarily to monetary incentives, others seem to view effort as a reciprocal response to being trusted with the task, in line with previous evidence from the laboratory (Fehr and Gächter, 1998). Finally, workers who failed an attention check or needed additional attempts to answer com-

prehension questions exert marginally less effort, consistent with relatively lower engagement in the study.

Taken together, the Round 1 evidence suggests that variation in initial effort is driven by workers' characteristics, rather than by the contextual feature of whether punishment is available. This finding may inform models in which the threat of punishment disciplines effort: in our context, merely introducing a punishment technology does not substantially reshape workers' initial labor-supply decisions on net. However, as we describe next, the actual experience of being punished has meaningful consequences for future effort provision.

4.2.2 Responses to Punishment

On average, workers increase effort in Round 2, completing about 4 more sliders ($p < 0.01$). Recall that, at the conclusion of Round 1, workers learn whether there was a good outcome and, if they were assigned to the Punishment treatment, their manager's punishment decision. In addition, workers learn whether or not their manager observes only the outcome for the round (Hidden Effort treatment) or the outcome and their exact number of sliders completed (Known Effort treatment). Table 5 displays the average increase in effort by treatment assignment for the full 2×2. Workers who learn that their manager has seen (and will see) the number of sliders that they complete increase their Round 2 effort by 1.2 more sliders ($p < 0.01$) than workers in the Hidden Effort treatment.

This section focuses on how Round 1 punishment decisions shape worker effort in Round 2. Table 5 shows that, on average, increases in effort are similar across the No Punishment and Punishment arms. However, as expected, this masks substantial heterogeneity, particularly in the Punishment treatment. To understand this fully, we need to consider not just the possibility of punishment, but the exact amount of punishment that is received. Intuitively, a worker who *could* have been punished by their manager may react much differently than a worker who received a large sanction, even holding Round 1 effort fixed.

Table 5: Effort Increases in Round 2 by Treatment

Average Increase in Number of Sliders Completed (Round 2 – Round 1)			
	Unknown Effort	Known Effort	
No Punishment	3.23 (0.55)	4.92 (0.36)	$p < 0.01$
Punishment	4.04 (0.50)	4.73 (0.43)	n.s.
	n.s.	n.s.	

Note: Each cell presents the average increase in effort from Round 1 to Round 2. Row ends present p-values from two-sided t-tests across the Unknown and Known Effort treatment, conditional on Punishment Treatment status. Column ends present p-values from two-sided t-tests across the No Punishment and Punishment treatment, conditional on Effort Treatment status.

Table 6 explores these effects, predicting a worker's Round 2 effort, broken out by the manager's sanction decision from Round 1. We also include controls for the worker's Round 1 effort, whether there was a good outcome in Round 1, and an indicator for the Known Effort treatment. In Columns 1 - 4, the key focus is on the indicator for the Punishment treatment, asking whether, conditional on a manager's Round 1 sanction amount, the experience of actually receiving that sanction impacts Round 2 effort. While the estimates are noisy given the data partition, we see the type of heterogeneity we would expect. Column 1 presents data for workers whose managers chose a sanction of 0, i.e., no punishment. Within this sub-group, being assigned to the Punishment treatment has a *positive* impact on future effort. That is, learning that you were not punished when you could have been increases effort, relative to a context where punishment was never a possibility. What happens as the sanction amount increases? Columns 2 - 4 show that for any non-zero sanction, receiving that sanction directionally reduces future effort, with directionally larger impacts as the sanction amount grows.

Table 6: How Round 2 Effort Responds to Round 1 Punishment Decisions

Round 2 Effort (Sliders Completed)					
	Manager's Round 1 Sanction Decision				
	$S = 0$	$S \in [1, 3]$	$S \in [4, 6]$	$S \geq 7$	Pooled
Punishment Arm	0.80*	0.43	-0.66	-2.84	0.92**
	(0.46)	(1.51)	(1.91)	(1.97)	(0.46)
R1 Sanction Amt.		0.070	-2.75	-1.93**	-0.17
		(1.10)	(2.70)	(0.80)	(0.13)
Pun. Arm × R1 Sanction Amt.					-0.41**
					(0.19)
R1 effort	0.87***	0.83***	0.59***	0.63***	0.81***
	(0.023)	(0.077)	(0.11)	(0.11)	(0.024)
Good outcome	0.10	-2.80*	-3.30	-5.90	0.050
	(0.46)	(1.65)	(3.25)	(5.89)	(0.46)
Known-effort arm	0.93**	2.46	1.08	1.83	1.32***
	(0.46)	(1.67)	(1.99)	(2.07)	(0.45)
Constant	7.28***	6.45**	25.7*	30.1***	8.57***
	(0.77)	(3.23)	(14.4)	(8.16)	(0.77)
Observations	1,140	140	159	135	1,574
Adjusted R^2	0.609	0.488	0.225	0.278	0.530

Note: OLS of Round 2 worker effort N_{g2} (sliders) across groups (one observation per group). Columns 1 - 4 split the sample according to Manager's Round 1 sanction decision. Column 5 uses all groups. All columns use heteroskedasticity-robust standard errors; standard errors displayed in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column 5 pieces these results into a coherent framework by pooling all workers. We predict Round 2 effort from the indicator for being in the Punishment treatment, the manager's Round 1 sanction choice, and their interaction. The interaction is the key coefficient

of interest, allowing us to ask how effort responds to increasingly larger sanctions in the Punishment treatment. We estimate a significant, negative coefficient of approximately -0.4: each additional token deducted in Round 1 lowers Round 2 effort by roughly 0.4 sliders, relative to the counterfactual where the same sanction is chosen but (randomly) not implemented. Our results show that a manager who has the power to punish, but refrains from doing so, elicits higher subsequent effort; a manager who exercises that power, especially at high levels, triggers a reduction in effort. These patterns are consistent with past work on potential backfire effects of fines or sanctions (Gneezy and Rustichini, 2000; Falk and Kosfeld, 2006) and the role of reciprocity in the workplace (Kube et al., 2012, 2013). In our setting, implemented punishment appears to at least somewhat crowd out workers' intrinsic or reciprocal motivation and reduce their willingness to exert effort. We also find that the response to an implemented sanction is the same above and below the median of Round-1 effort (Table A31). Thus, this result is not driven by workers who had already exerted high effort, as one would expect if the reduction reflected a sense of having been punished unfairly.

Of particular interest in our setting is whether these effects vary by race or gender. Table 7 presents the results of running the specification of Column 5 of Table 6 separately by race and then by gender. Our results reveal that male workers, particularly white male workers, drive the patterns we observed in Table 6. Column 1 shows responses to punishment for white workers. The interaction between the punishment treatment and the sanction amount is significantly negative (around -0.61): each additional token deducted in Round 1 reduces Round 2 effort by more than half a slider relative to the No Punishment counterfactual. For Black workers, by contrast, the interaction between the punishment arm and the sanction amount is considerably smaller in magnitude and not statistically significant (see Column 2). The gender difference is also stark. Men (Column 3) exhibit a strong negative response to the level of implemented punishment (the punishment-arm $\times$ sanction interaction is approximately -0.8 and statistically significant), whereas women's effort is only weakly and

imprecisely related to the sanction intensity (Column 4).[15]

[15]The pooled regression is in Table A32 in the Appendix.

Table 7: Responses to Punishment by Race and Gender

	Round 2 Effort (Sliders Completed)			
	White	Black	Men	Women
Punishment Arm	1.32*	0.50	1.91**	0.21
	(0.68)	(0.59)	(0.77)	(0.55)
R1 Sanction Amt.	-0.28	-0.051	-0.014	-0.29*
	(0.20)	(0.16)	(0.22)	(0.15)
Pun. Arm × Sanction Amt.	-0.61*	-0.29	-0.82**	-0.12
	(0.33)	(0.24)	(0.34)	(0.22)
R1 effort	0.82***	0.83***	0.83***	0.82***
	(0.036)	(0.030)	(0.037)	(0.030)
Good outcome	-0.36	0.46	-0.55	0.57
	(0.69)	(0.60)	(0.80)	(0.55)
Known-effort arm	2.28***	0.24	1.80**	0.98*
	(0.70)	(0.57)	(0.75)	(0.56)
Male worker	-1.47**	0.037		
	(0.72)	(0.57)		
Black worker			2.27***	0.61
			(0.76)	(0.55)
Constant	7.98***	8.97***	6.13***	8.60***
	(1.28)	(1.01)	(1.40)	(1.04)
Observations	785	789	644	930
Adjusted R^2	0.500	0.568	0.518	0.547

Note: OLS of Round 2 worker effort N_{g2} (sliders) across groups (one observation per group). Columns 1 and 2 present the results for white and Black workers, respectively. Columns 3 and 4 present the results for men and women, respectively. All columns use heteroskedasticity-robust standard errors; standard errors are displayed in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

These results indicate that "backlash" to punishment is not uniform. It is particularly strong for white men, who respond both more positively to being spared and more negatively to being punished. Women and Black workers, in contrast, adjust their effort less in response to the same sanctions. While we can only speculate on exact mechanisms, two possibilities seem worth noting. First, punishment may be a larger negative surprise for groups more likely to expect favorable treatment, generating larger effort responses from white men. Alternatively, these differences may stem from differences in future expectations. When deciding whether or not to reduce effort in response to Round 1 punishment, workers may consider how much punishment they expect to receive after Round 2, with white men perhaps expecting less retaliation for lower effort.

4.2.3 Punishment and the decision to quit

Beyond adjusting intensive-margin effort, workers in our study can also respond to punishment by exiting the workplace. After completing Round 2, but before learning the respective outcome, they choose whether to opt in to Round 3. Playing Round 3 is weakly payoff-increasing for both the worker and the manager, as it offers additional expected earnings in exchange for a modest time cost. In addition, managers only learn of the possibility of the third round if their worker opts in. Thus, a worker who opts out both forgoes potential earnings and withholds an opportunity from the manager.

Overall, about 65% of workers choose to play Round 3. Average opt-in rates do not differ significantly across the Punishment and No Punishment treatments, mirroring the null average effect of the punishment regime on Round 2 effort. However, once again, realized sanctions matter. Table 8 reports linear-probability models in which the dependent variable is an indicator for choosing to play Round 3. To minimize endogeneity concerns and parallel the analysis of Table 6, we condition only on Round 1 realizations, including Round 1 effort, whether or not there was a good outcome in Round 1, and an indicator for the Known-Effort treatment.

Table 8: Willingness to Continue Working for Manager in Round 3

Opted-in to Round 3 with Same Manager					
	All	White	Black	Men	Women
Punishment Arm	0.048*	0.059	0.033	0.019	0.071**
	(0.027)	(0.037)	(0.038)	(0.041)	(0.035)
R1 Sanction Amt.	0.0016	0.0079	-0.0050	-0.0069	0.0081
	(0.0061)	(0.0086)	(0.0087)	(0.0094)	(0.0079)
Pun. Arm × Sanction Amt.	-0.028***	-0.038***	-0.020*	-0.0050	-0.046***
	(0.0084)	(0.012)	(0.011)	(0.013)	(0.011)
Effort in Round 1	0.0055***	0.0058***	0.0052***	0.0047***	0.0062***
	(0.0011)	(0.0015)	(0.0016)	(0.0016)	(0.0015)
Good Outcome in Round 1	0.038	0.084**	-0.012	0.090**	0.0017
	(0.027)	(0.037)	(0.039)	(0.041)	(0.035)
Known Effort	0.024	-0.0066	0.057*	0.071*	-0.0085
	(0.024)	(0.033)	(0.034)	(0.037)	(0.031)
Male	0.022	-0.0057	0.049		
	(0.024)	(0.034)	(0.034)		
Black	0.018			0.045	0.0019
	(0.024)			(0.039)	(0.031)
Constant	0.45***	0.45***	0.47***	0.45***	0.46***
	(0.042)	(0.056)	(0.054)	(0.063)	(0.055)
Observations	1,574	785	789	644	930
Adjusted R^2	0.033	0.044	0.024	0.027	0.043

Note: OLS of worker decision to opt-in to Round 3 with the same manager (one observation per group). Column 1 presents all groups. Columns 2 and 3 present the results for white and Black workers, respectively. Columns 4 and 5 present the results for men and women, respectively. All columns use heteroskedasticity-robust standard errors; standard errors displayed in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column 1 displays the results for all workers. First, among workers in the punishment arm whose manager chose a zero sanction, the probability of opting in is about 5 percentage points higher than for comparable workers in the no-punishment arm ($p < 0.10$), echoing the positive effect of avoiding Round 1 punishment on Round 2 effort. Second, for workers in the punishment arm who do receive a Round 1 sanction, each additional token deducted in Round 1 reduces the probability of opting in to Round 3 by around 3pp relative to the No Punishment counterfactual ($p < 0.01$). Intuitively, higher realized punishment not only discourages effort in the next period but also increases the likelihood that workers exit the relationship altogether, even though doing so is privately and socially costly.

Columns 2 and 3 present the results by race, splitting the sample into white workers (Column 2) and Black workers (Column 3). We observe that for both sub-samples, there is a negative impact of realized punishment on willingness to play Round 3, with effects between 2 - 4pp. Columns 4 and 5 reveal that these effects are driven entirely by women. Women's willingness to engage in Round 3 depends strongly on their manager's punishment decisions. In particular, women in the Punishment treatment who were not punished are approximately 7pp more likely to opt in to Round 3 than their No Punishment treatment counterparts. Each additional token of punishment reduces the likelihood of women playing Round 3 with their manager by nearly 5pp ($p < 0.01$) but has little effect on men's opt-in decisions on average. These patterns are especially stark for White women: in the punishment arm, their opt-in rate falls from roughly 73% when they are not punished to about 44% when they are punished in Round 1, a drop of nearly 30pp ($p < 0.01$). A placebo comparison using the No Punishment treatment, where managers make identical Round 1 sanction decisions but the decisions are not implemented, shows no corresponding difference, highlighting that it is the experience of being punished that seems to matter. These high rates of exit among women in the Punishment treatment are noteworthy, given that workers have the option to engage in Round 3 but simply "shirk," yielding themselves higher monetary payoffs (even net of a sanction) but at the possible additional psychological cost of being punished or perceived

negatively by their manager.

Recall that it was men, in particular white men, who most sharply adjusted Round 2 effort on the intensive margin in response to Round 1 punishment. Interestingly, when we shift to extensive margin reactions, it is women, not men, who are adjusting most sharply. Thus, in the context of a continuing relationship, men appear more willing to shirk in response to perceived poor treatment, risking additional punishment or reputational costs; but, when given the possibility to terminate a relationship, women appear more willing to take this (costly but quiet) option.

One concern is that punishment could be reducing workers' willingness to participate in the experiment more broadly, for instance by lowering expected earnings or diminishing other-regarding preferences toward generic partners. To address this, we exploit a follow-up choice in which workers indicate whether they would be willing to play Round 3 with a new, randomly-assigned manager. Table 9 shows that punishment has essentially no effect on the probability of opting in to Round 3 *with any manager*: the main and interaction coefficients for the punishment arm and the sanction amount are small and statistically insignificant for the full sample and for most subgroups. Instead, punishment specifically reduces willingness to continue working with the original manager. In the Punishment treatment, the share of workers who opt in with their existing manager falls from about 70% when they were not punished to roughly 50% when they were punished, while total willingness to play with some manager remains around 70% regardless of punishment status.

Workers' responses are also the same whether their manager is a woman or a man(Appendix Table A14, Panels C and D).

Table 9: Willingness to Continue Working for ANY Manager in Round 3

Opted-in to Round 3 with Any Manager					
	All	White	Black	Men	Women
Punishment Arm	0.046*	0.062*	0.028	0.0021	0.081**
	(0.026)	(0.036)	(0.038)	(0.040)	(0.034)
R1 Sanction Amt.	-0.00020	0.0027	-0.0036	-0.0076	0.0058
	(0.0061)	(0.0086)	(0.0086)	(0.0093)	(0.0079)
Pun. Arm × Sanction Amt.	-0.0024	-0.0049	0.00061	0.019	-0.019*
	(0.0082)	(0.012)	(0.011)	(0.012)	(0.011)
Effort in Round 1	0.0052***	0.0057***	0.0046***	0.0042***	0.0061***
	(0.0011)	(0.0015)	(0.0016)	(0.0016)	(0.0015)
Good Outcome in Round 1	0.023	0.055	-0.011	0.092**	-0.027
	(0.026)	(0.036)	(0.038)	(0.039)	(0.035)
Known Effort	0.023	0.0035	0.043	0.078**	-0.017
	(0.023)	(0.033)	(0.034)	(0.036)	(0.031)
Male	0.023	0.013	0.031		
	(0.024)	(0.033)	(0.034)		
Black	0.0038			0.012	0.0024
	(0.024)			(0.037)	(0.031)
Constant	0.49***	0.47***	0.52***	0.50***	0.49***
	(0.042)	(0.055)	(0.054)	(0.062)	(0.054)
Observations	1,574	785	789	644	930
Adjusted R^2	0.016	0.023	0.006	0.026	0.018

Note: OLS of worker decision to opt-in to Round 3 with **any** manager, i.e., with the same manager or with a new, randomly-assigned one (one observation per group). Column 1 presents all groups. Columns 2 and 3 present the results for white and Black workers, respectively. Columns 4 and 5 present the results for men and women, respectively. All columns use heteroskedasticity-robust standard errors; standard errors displayed in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3 Group-Level Analysis: Impacts of Punishment on Earnings and Efficiency

Having studied managers' and workers' decisions in isolation, we now ask what punishment does to the pair as a whole. For each group and round, we look at three outcomes: (i) Joint surplus, the total tokens the pair gains; (ii) Welfare, which nets out from the joint surplus the worker's cost of effort; and (iii) Share, the manager's share of the total surplus. We compare the Punishment and No-Punishment arms, where, in the first round, managers face identical decisions, but only in one is the sanction implemented. We ask three questions. First, does punishment increase or reduce the pie? Then, who bears the difference? And finally, does observing effort change these answers?

4.3.1 Punishment reduces the pie without value creation

In Table A33, we look at both round 1 and 2 payoffs, with round fixed effects, and observe that punishment makes the pair worse off in both rounds. Relative to the No-Punishment treatment arms, the ability to punish lowers a group's joint surplus by 1.3 tokens per round ($p < 0.01$). The loss falls entirely on the worker, whose payoff drops by 1.5 tokens, and the manager's own payoff is statistically unchanged (0.24, not statistically significant). Punishment falls only on the worker; the manager receives no direct reward, and it does not buy enough effort to offset the value it destroys. If punishment raised effort, the welfare loss would exceed the surplus loss, because welfare also subtracts the worker's cost of effort. It does not (-1.19 vs. -1.31 tokens). Note that exerting effort is socially optimal in this setting. Each slider raises the probability of a good outcome by one percentage point, worth 1 cent in expectation to the worker alone. For 57% of workers, the elicited cost of completing one more slider is below this 1-cent return, and the median cost is 0.83 cents.[16] The elicited cost is also likely an upper bound on the cost of effort during the task. In the price-list

[16]If we exclude those who rejected all offers, then the median cost per slider is 0.67 cents and 62.4% of workers have a cost below 1.

exercise, each extra slider costs both effort and time. In the task itself, time is not at stake because every worker waits the full 90 seconds regardless of how many sliders they complete, so the only cost of an additional slider is the effort it takes.

If we include Round 3 in the analysis, the surplus loss is smaller due to increases in managers' pay in Round 3, which is consistent with selection into playing a third round with the same manager (Table A34).

Cumulating surplus across all three rounds, among the positively selected groups that chose to continue to Round 3, punishment still lowers joint surplus by about 2.6 tokens, borne almost entirely by the worker. The point estimate is larger in the Known-Effort arm, by 4.8 tokens, but the two arms are statistically distinguishable only on the welfare measure ($p < 0.05$), not on surplus. The per-round cost is smaller here than in the full two-round sample precisely because the groups most damaged by punishment are the ones whose workers opt out of Round 3, so these figures are a lower bound (Table A35).

The fines actually implemented throughout the experiment destroyed roughly 2500 tokens, about \$250, or 12% of the potential group surplus in the Punishment arm. They led to no measurable increase in output. The damage is broadly even-handed across workers: splitting the sample by worker race and gender, every subgroup loses between 1.3 and 1.8 tokens of surplus, with no statistically significant differences across groups (Table A36). Despite the documented punishment gap, the efficiency cost of punishment itself is a general feature of the institution.[17]

In summary, punishment destroys value, and disciplined effort offers no offsetting efficiency gain.

[17]The point estimate of the surplus loss is slightly larger for groups with a Black worker (-1.8 vs. -1.3 for a White worker), consistent with those workers being punished more, but the difference is not statistically significant.

4.3.2 Distributional consequences: relative gains and added risk

Because workers bear all the losses, managers' share of the total pie increases, but without an absolute gain. Punishment also makes group outcomes more unequal and more precarious. It lowers average surplus and increases its variability, increasing the share of zero-surplus outcomes. The standard deviation of joint surplus rises from 4.7 tokens in the No-Punishment arm to 6.4 under punishment, and the share of groups ending a round with zero-surplus rises from zero to around 8.5% (Table A37). Additionally, under punishment, revealing effort lowers the share of group-rounds at the maximum surplus of 20 by around 4 pp ($p = 0.085$); without punishment, there is no such drop. This motivates us to study the role of information on efficiency.

4.3.3 Neither observation nor time rescues punishment

A natural conjecture is that punishment is inefficient only because managers lack full information. Indeed, when managers observe effort, they target sanctions at low-effort workers (Table A38). However, this does not make it more efficient. Learning that their performance is observed makes workers exert more effort in the second round. However, it does so through information rather than punishment. In fact, punishment pushes in the opposite direction (Table A39). If anything, combining punishment with observing effort seems to be a more damaging institutional setting, although the difference is overall not statistically significant (Table A40).

Taken together, the group-level evidence is unambiguous: in this setting the power to punish is close to a pure loss. It shrinks the surplus the pair produces, assigns the cost entirely to the worker, adds a tail of zero surplus, and neither observing effort nor repeating the interaction turns it to the group's advantage. Alongside the finding that punishment falls disproportionately on low-effort Black men, this makes the institution not only inefficient but inequitable.

5 Conclusion

In this paper, we study the incidence and impact of punishment in the context of a work relationship. Our controlled experiment varies exogenously whether managers' chosen sanction is implemented, and whether they observe the worker's effort prior to making the decision. Our setting allows us to study the downstream consequences for labor supply on the intensive and extensive margin, with particular attention to differences by race and gender. These questions are challenging to study in observational data, but have important implications for understanding the full costs and benefits of punishment at work.

Focusing on race, we observe that managers do not treat Black and white workers differently on average. But, we see important race disparities among workers observed to contribute relatively low effort. We find that managers sanction Black workers who did no sliders 1.3 tokens more than white workers that did no sliders. The gap comes mostly through the extensive margin, on whether to sanction or not, and it shrinks as effort rises, closing completely near the average effort level. Managers' stated expectations and standards are the same for Black and white workers, and the gap does not appear when effort is hidden, which is where a race-based inference about effort should matter most. What differs, then, is the response to identical information. We do not find this difference once managers have a second round of information about the worker.

Gender plays a different role. Managers do not punish women and men differently, at any level of effort. Gender matters instead for how workers react. Knowing that the manager can punish does not change first-round effort for any group. Being punished does. White men respond most strongly in effort, working harder when spared and less hard when sanctioned. Women and Black workers change their effort much less. Women, however, are the ones who walk away when given the opportunity: after a sanction, their willingness to continue at all with the same manager drops sharply, from about 73% to 44% among white women, while their willingness to work with a new manager does not change.

In this environment, punishment harms everyone. Beyond reducing workers' surplus, it

does not increase effort and so fails to increase managers' payoffs. Our results on labor supply further suggest that punishment may distort the talent pool, driving women in particular away from working relationships that feature punishment.

Our results point to three lessons about punishment in a working relationship. First, the threat of punishment and its use are very different instruments: the threat alone leaves labor supply unchanged, while using punishment changes it, mostly for the worse. Second, workers respond to punishment differently. White men withdraw effort while staying in the relationship, whereas white women keep working as before but leave when they can. Third, biases in who is punished do not come from either statistical discrimination or purely taste-based discrimination. In our setting, they appear only when a manager has observed poor performance and must judge it.

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A Appendix

A.1 Matching Algorithm

Because the experiment requires two participants to play simultaneously, assignment depends on the pool of participants available at each point in time. When a participant selects a dark skin tone, they are automatically assigned the role of the worker and placed into a mixed-group. Their arrival also increments a "mixed" queue counter, recording that a Black worker is now waiting for a white manager. When a participant selects the pale skin tone, the assignment depends on the current state of that queue — that is, on who has arrived before them. If at least one Black worker is currently queued, the pale participant is assigned the manager role in a mixed group. If no Black worker is queued, the pale participant is

assigned to a same-race group and to a role within that group. Once a group is complete (worker and manager), they receive a treatment number drawn from a counter that cycles through the four treatment arms in sequence and considers the balance of group types within each treatment arm. If no complementary partner arrives before the matching timeout (3 minutes), the participant is recorded as unmatched, and participation ends.

Given the low number of Black individuals in Prolific, the later stage of recruitment, and the need to ensure half of the groups were mixed-race, we changed the allocation algorithm to prioritize mixed-race groups. A same-race pair is formed only under more restrictive conditions: at least two light-skinned participants must be waiting, and the first (i.e., longest-waiting, earliest-arrived) of them must have been waiting at least 2.5 minutes, and a quota condition on the ratio of all-white to mixed pairs must be satisfied. When these conditions hold, the two waiting white participants are paired, with the earliest-arrived assigned the role of worker and the next the role of manager, and both given the same treatment from a separate cycling counter. Any participant who remains unmatched beyond the overall matching timeout is recorded as unmatched. In this version, then, arrival order governs matching in real time — mixed pairs are formed as soon as the two required skin tones are jointly present, and all-white pairs are formed only after the earliest waiting white participant has waited a set interval and the quota permits.

A.2 Reasoning

All managers answer about Round 1 decisions, and only managers in the punishment treatment arms answer these questions about other rounds. Trivially, only managers in the punishment arm whose worker chose to play a third round answer the corresponding question. We asked a Large Language Model (LLM) to analyze these reasons blindly and classify them into categories. They are the following: Effort Evaluation (based on the manager's evaluation of the worker's effort/performance), Fairness (based on the manager's sense of fairness / a just outcome), Personal Gain (the manager's own potential gain or loss), Outcome-based

(based on the task outcome, regardless of effort), Motivation (desire to motivate the worker to do better in the future), Unwarranted (the manager saw no reason or need to fine the worker), Empathy (understanding or relating to the worker's feelings and situation), Lack Of Control (recognition that the worker had limited control over the outcome), Arbitrary (the amount was chosen at random or without a stated reason), Trust (level of trust in the worker / giving them the benefit of the doubt), and Punishment as Incentive (belief that punishment itself can serve as an incentive to improve).[18] Given the similarity between Motivation and Punishment as an incentive, we combine those two reasons in the analysis. Importantly, each reason is assigned to one category. The two most common justifications are Effort Evaluation and Fairness, followed by Personal gain and Outcome-based (see Table A20).

Table A21 shows that managers' justifications align with their behavior.[19] Those who motivate sanctioning with effort-based evaluation, the outcome they received, and the use of punishment as a tool to motivate are the ones who implement larger sanctions. Those who justify their punishment decisions with empathy for the worker, lack of control, and unwarranted punishment impose lower sanctions. They also seem to use the information they receive to justify their decisions. Compared to the HE arm, when managers observe effort, they are 25pp ($p < 0.01$) and 2.7pp ($p < 0.10$) more likely to use effort- and motivation-based reasoning, respectively, in detriment of most of the other reasonings, as shown in Table A22.

If instead we restrict the analysis to where the racial gap is found (managers in the

[18]The taxonomy was built in two steps. First, a model (Llama 3.3 70B) read a random sample of 200 Round-1 explanations with no predefined list and proposed its own categories with definitions. This produced the nine categories other than Unwarranted and Arbitrary. Second, on reading the labeled statements, we found that explanations giving no reason at all ("it was a wild guess") had been forced into the closest substantive category, most often Lack of Control. We therefore added a tenth category for such statements. Inspection of that category showed it mixed two different things, managers who saw no reason to fine and managers who gave no reason for the amount, so we split it into Unwarranted and Arbitrary. All explanations from all rounds were then classified into the final eleven categories with a single model (gpt-oss-120b, used because the original model's endpoint had been retired) and a single prompt. The model never sees any participant identifier.

[19]The reference category in the table is Personal gain. We dropped trust from the analysis due to the small sample size (28).

Known-Effort arm matched with a male worker), evaluating the gap at low effort, clear patterns arise. We observe a racial gap among managers who judged the worker's effort or the outcome. For managers who cite fairness, motivation, personal gain, or who saw no reason to fine, we observe no gap. The few who cite empathy show a reversed gap (Table A23). In sum, managers who frame their decision as an assessment of past or future performance sanction a Black worker who completed no sliders by 3.7 tokens more than a White worker, and the gap closes by 0.13 tokens per slider. Managers giving any other reason show no gap (Table A24). This pattern is absent among female workers, for whom there is no racial gap to begin with.

We also run a simple word-count exercise to examine managers' justifications in depth. We categorize justifications into three ideas: blame, encouragement, and softening, capturing the lazy-Black worker stereotype, the effectiveness of punishment, and the benefit of the doubt due to outcome uncertainty, respectively. Then, within each, we counted for the presence of a short list of words: blame (lazy, nothing, unacceptable, deserve, didn't try, no effort), encouragement (encourage, motivate, hope, next round, do better), and softening (fair, first try, not his fault, felt bad, he tried). A justification counts as using an idea if it contains at least one word from that list. We then compare the share of justifications that use each idea for Black and White workers, with a test of proportions. Again focusing on the population where the racial gap exists, we find suggestive evidence that blame language is more common for Black workers, and encouragement and softening words are more common for white workers (See Table A25). We are underpowered to detect consistent statistical significance, but the point estimates are meaningful, especially when the worker exerts less effort.

In sum, the stated reasons locate the racial gap at one point: managers who judge a Black man's low effort as a performance failure fine him more than a white man with the same record, while managers who reason from fairness, empathy, or their own payoff treat the two alike. The justifications themselves differ in the same direction, with more character

language for Black workers and more excuses and encouragement for white workers.

A.3 Tables and Figures

Table A1: Avatar's skin tone and race/ethnicity

	Chosen avatar skin tone			
Self-reported race	Dark brown	Brown	Pale	*N*
Black	34.6	61.9	3.5	740
White	0.8	2.2	96.9	2393
Other race	0.0	13.3	86.7	15
All	8.8	16.3	74.9	3148

Note: This table shows the rate at which each avatar skin tone is chosen within a given self-reported race.

Table A2: Sample

Stage	*N*	Share of started
Started the study (completed intro demographics)	6334	1.000
Matched into a manager–worker pair	4873	0.769
Completed all rounds (no own/partner dropout)	3148	0.497
Passed attention checks (≤1 failed)	3148	0.497
Analysis sample	3148	
Managers	1574	
Workers	1574	
Excluded (of 6334 started):		
Never matched to a partner	1461	0.231
Own dropout	909	0.144
Partner dropped out	816	0.129

Note: Sample construction from all participants who started the study to the analysis sample. The excluded rows are mutually exclusive and sum to the difference between the number who started and the analysis sample; "partner dropped out" includes partners who left after the interactive part of the study.

Table A3: Workers and Managers' Characteristics

	Managers (all)	Workers (all)
Demographics		
Age	43.36	40.71
	(13.32)	(13.35)
Male	0.386	0.409
Black, self-reported	0.013	0.457
White, self-reported	0.982	0.538
High school or less	0.218	0.185
Currently employed	0.748	0.756
Elicited preferences (0–10 scales)		
Risk tolerance	3.25	3.52
	(1.00)	(0.94)
Patience	3.93	3.98
	(0.76)	(0.74)
Altruism	4.04	4.11
	(0.80)	(0.75)
Trust	3.13	3.03
	(1.00)	(1.02)
Positive reciprocity	4.39	4.38
	(0.59)	(0.62)
Negative reciprocity	1.96	2.10
	(0.93)	(0.99)
Task measure		
Effort cost (MPL, cents/slider)	1.03	0.93
	(0.82)	(0.77)
Data quality		
Attention checks failed	0.37	0.40
	(0.48)	(0.49)
Comprehension mistakes	0.82	0.59
	(1.08)	(0.80)
Observations	1574	1574

Note: Baseline characteristics of the analysis sample (one observation per participant), by role, measured in the intro survey before role and treatment assignment. Continuous variables report the mean with the standard deviation in parentheses below; binary variables report the proportion.

Table A4: Workers' Characteristics by Race and Gender

	By race			By gender		
	White	Black	(B−W)	Female	Male	(M−F)
Demographics						
Age	43.92	37.52	-6.41***	40.78	40.61	-0.17
	(13.64)	(12.25)		(13.56)	(13.04)	
High school or less	0.214	0.156	-0.058***	0.182	0.189	0.008
Currently employed	0.720	0.792	0.072***	0.706	0.828	0.121***
Elicited preferences (0–10 scales)						
Risk tolerance	3.30	3.75	0.46***	3.42	3.68	0.27***
	(0.97)	(0.85)		(0.96)	(0.88)	
Patience	3.96	3.99	0.03	3.92	4.07	0.15***
	(0.69)	(0.79)		(0.74)	(0.74)	
Altruism	4.04	4.17	0.14***	4.15	4.05	-0.10***
	(0.77)	(0.73)		(0.75)	(0.76)	
Trust	3.15	2.92	-0.23***	3.02	3.06	0.04
	(0.97)	(1.05)		(1.02)	(1.01)	
Positive reciprocity	4.39	4.38	-0.01	4.41	4.34	-0.07**
	(0.59)	(0.65)		(0.59)	(0.66)	
Negative reciprocity	2.00	2.20	0.20***	2.02	2.21	0.19***
	(0.96)	(1.01)		(0.98)	(0.99)	
Task measure						
Effort cost (MPL, cents/slider)	0.97	0.88	-0.09**	0.92	0.93	0.01
	(0.74)	(0.79)		(0.78)	(0.75)	
Data quality						
Attention checks failed	0.36	0.44	0.08***	0.38	0.42	0.03
	(0.48)	(0.50)		(0.49)	(0.49)	
Comprehension mistakes	0.38	0.80	0.42***	0.54	0.66	0.12***
	(0.76)	(0.79)		(0.74)	(0.88)	
Observations	785	789		930	644	

Note: Workers' baseline characteristics of the analysis sample (one observation per participant), separate measured in the intro survey before role and treatment assignment. Continuous variables report the mean with the standard deviation in parentheses below; binary variables report the proportion. *Diff. (B− W)* and *Diff. (M−F)* are the black−white and male−female difference, respectively. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Balance across treatment arms

	Treatment arm				Joint
	NP×HE	NP×KE	P×HE	P×KE	p
Age	41.78	41.93	42.02	42.42	0.809
Male	0.395	0.415	0.407	0.371	0.343
Black, self-rep.	0.230	0.240	0.232	0.238	0.935
High school or less	0.209	0.207	0.187	0.202	0.659
Currently employed	0.740	0.757	0.762	0.748	0.747
Risk tolerance	3.38	3.31	3.42	3.43	0.059
Patience	3.90	3.97	3.99	3.94	0.098
Altruism	4.06	4.06	4.10	4.07	0.687
Trust	3.08	3.06	3.06	3.12	0.564
Positive reciprocity	4.40	4.38	4.40	4.37	0.620
Negative reciprocity	2.01	2.01	2.07	2.03	0.665
Effort cost (MPL)	0.98	0.97	0.99	0.98	0.953
Mixed-race pair	0.495	0.517	0.486	0.506	0.828
Observations	784	820	766	778	

Note: Mean of each baseline characteristic by treatment arm (NP: No punishment, P: Punishment; HE: Hidden effort, KE: Known effort), analysis sample, one observation per participant. Binary variables report the proportion. The *Joint p* is from a test that the four-arm means are equal, obtained by regressing the characteristic on the treatment-arm indicators with standard errors clustered by pair. *Mixed-race pair* is the share of participants assigned to a mixed-race group, a check that the real-time matching algorithm did not induce imbalance. The final row reports the number of observations per arm.

Table A6: Balance of Baseline Characteristics: Managers

	Treatment arm (Punishment × Known effort)				
	NP×HE	NP×KE	P×HE	P×KE	p
Male	0.36	0.40	0.41	0.38	0.515
	(0.48)	(0.49)	(0.49)	(0.49)	
Age	43.57	43.97	43.47	42.41	0.363
	(13.69)	(13.47)	(13.54)	(12.56)	
High school or less	0.23	0.21	0.21	0.22	0.881
	(0.42)	(0.41)	(0.41)	(0.42)	
Employed	0.74	0.75	0.76	0.74	0.813
	(0.44)	(0.43)	(0.43)	(0.44)	
US resident	0.96	0.96	0.96	0.96	0.952
	(0.19)	(0.21)	(0.19)	(0.19)	
Risk taking	3.19	3.21	3.32	3.27	0.222
	(1.01)	(0.98)	(0.99)	(1.01)	
Patience	3.93	3.90	3.99	3.89	0.233
	(0.75)	(0.75)	(0.72)	(0.82)	
Altruism	4.05	4.05	4.04	4.03	0.980
	(0.80)	(0.80)	(0.77)	(0.82)	
Trust	3.09	3.16	3.08	3.17	0.439
	(0.97)	(0.99)	(1.04)	(0.98)	
Good intentions	2.90	3.00	2.96	3.00	0.392
	(0.99)	(1.04)	(0.99)	(1.02)	
Positive reciprocity	4.40	4.38	4.43	4.34	0.191
	(0.58)	(0.58)	(0.58)	(0.62)	
Negative reciprocity	1.92	1.96	2.00	1.97	0.668
	(0.96)	(0.94)	(0.93)	(0.91)	
Effort cost (per slider)	1.00	1.00	1.10	1.02	0.293
	(0.84)	(0.80)	(0.81)	(0.85)	
Observations	392	410	383	389	

Note: Each cell reports the mean of a baseline (pre-treatment) characteristic, with the standard deviation in parentheses below. Baseline characteristics are measured in the intro survey, before role and treatment assignment. The columns report means across the four treatment arms (NP: No punishment, P: Punishment; HE: Hidden effort, KE: Known effort); the accompanying p-value is from a joint F-test that the four-arm means are equal, obtained by regressing the characteristic on the treatment-arm indicators with robust standard errors. The final row reports the number of observations.

Table A7: Balance of Baseline Characteristics: Workers

	Treatment arm (Punishment × Known effort)				
	NP×HE	NP×KE	P×HE	P×KE	p
Male	0.43	0.43	0.41	0.36	0.108
	(0.50)	(0.50)	(0.49)	(0.48)	
Age	39.99	39.89	40.58	42.43	0.038
	(13.26)	(13.16)	(12.62)	(14.20)	
High school or less	0.19	0.20	0.17	0.18	0.682
	(0.39)	(0.40)	(0.37)	(0.39)	
Employed	0.74	0.76	0.76	0.76	0.911
	(0.44)	(0.43)	(0.43)	(0.43)	
US resident	0.96	0.96	0.96	0.96	0.941
	(0.19)	(0.21)	(0.19)	(0.19)	
Risk taking	3.58	3.41	3.52	3.59	0.024
	(0.95)	(0.96)	(0.93)	(0.91)	
Patience	3.88	4.04	3.99	3.99	0.027
	(0.79)	(0.69)	(0.72)	(0.77)	
Altruism	4.07	4.07	4.17	4.11	0.233
	(0.78)	(0.73)	(0.75)	(0.75)	
Trust	3.06	2.97	3.04	3.07	0.462
	(1.04)	(1.00)	(1.02)	(1.02)	
Good intentions	2.87	2.82	2.92	2.90	0.519
	(1.05)	(1.04)	(1.02)	(1.02)	
Positive reciprocity	4.40	4.38	4.37	4.39	0.956
	(0.65)	(0.61)	(0.61)	(0.62)	
Negative reciprocity	2.10	2.07	2.13	2.10	0.854
	(1.01)	(0.96)	(0.96)	(1.03)	
Effort cost (per slider)	0.97	0.93	0.88	0.93	0.453
	(0.76)	(0.74)	(0.77)	(0.80)	
Observations	392	410	383	389	

Note: Each cell reports the mean of a baseline (pre-treatment) characteristic, with the standard deviation in parentheses below. Baseline characteristics are measured in the intro survey, before role and treatment assignment. The columns report means across the four treatment arms (NP: No punishment, P: Punishment; HE: Hidden effort, KE: Known effort); the accompanying p-value is from a joint F-test that the four arm means are equal, obtained by regressing the characteristic on the treatment-arm indicators with robust standard errors. The final row reports the number of observations.

Table A8: Sample Attrition

	Treatment arm				Joint
	NP×HE	NP×KE	P×HE	P×KE	p
Attrition rate	0.353	0.295	0.363	0.339	0.055
Partner-dropout rate	0.163	0.140	0.171	0.161	0.124

Note: Attrition among matched participants (those assigned to a pair), by treatment arm (NP: No punishment, P: Punishment; HE: Hidden effort, KE: Known effort). *Attrition rate* is the share who did not complete the study; *Partner-dropout rate* is the share lost because their partner dropped out. The *Joint p* tests equality of the rate across arms (regression on the treatment-arm indicators, standard errors clustered by pair).

Table A9: Sample Attrition

	Retained	Dropped	Diff. p
Age	42.04	41.91	0.710
Male	0.397	0.393	0.700
Black, self-rep.	0.235	0.279	0.000
White, self-rep.	0.760	0.714	0.000
High school or less	0.201	0.215	0.193
Currently employed	0.752	0.736	0.148

Note: Baseline characteristics of retained (analysis-sample) versus dropped participants, measured in the intro survey. Binary variables report the proportion. *Diff. p* is from a heteroskedasticity-robust regression of the characteristic on a retained indicator. Dropped participants include those never matched to a partner and those who (or whose partner) dropped out.

Table A10: Summary Statistics: Punishment Decisions

Treatment arm	N	Mean	SD	% punishing	Mean | punished
No punishment × Hidden effort	392	1.27	2.59	25.3	5.01
No punishment × Known effort	410	1.70	3.07	31.0	5.47
Punishment × Hidden effort	383	1.36	2.96	21.7	6.25
Punishment × Known effort	389	1.70	3.01	32.1	5.28
All managers	1574	1.51	2.92	27.6	5.46

Note: This table shows the summary statistics of punishment decisions by treatment arms. N represents the number of observations, Mean is the average sanction managers chose, SD the standard deviation, and % punishing the percentage of managers who chose a strictly positive sanction. Mean | punished is the average sanction among managers who chose a strictly positive sanction.

Table A11: Unconditional Race and Gender Gaps in Punishment

Worker group	N	Mean sanction	SD	% punishing
By worker race				
White worker	785	1.35	2.77	25.4
Black worker	789	1.66	3.05	29.8
Difference (Black − White)		0.30**		4.4**
By worker gender				
Female worker	930	1.49	2.88	27.7
Male worker	644	1.53	2.98	27.3
Difference (Male − Female)		0.04		-0.4

Note: This table reports unconditional Round-1 punishment by worker race and by worker gender, one observation per manager. N is the number of managers, Mean sanction is the average number of tokens deducted, SD is its standard deviation, and % punishing is the share who choose a strictly positive sanction. Difference rows report the Black−White and Male−Female gap in the mean sanction and in the share punishing (percentage points); asterisks denote statistical significance from a heteroskedasticity-robust regression of the sanction on the single group indicator: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Punishment Decisions: Effort, Gender and Race

	S_{g1}			
	Women	Men	Black	White
Black worker	0.595	2.437***		
	(0.762)	(0.929)		
Male worker			1.031	-0.134
			(0.875)	(0.823)
Good outcome (G_{g1})	-1.851***	-1.580***	-1.669***	-1.822***
	(0.302)	(0.383)	(0.337)	(0.295)
Black × Good outcome	0.408	-0.186		
	(0.449)	(0.588)		
Male × Good outcome			-0.092	0.092
			(0.538)	(0.469)
Effort (N_{g1})	-0.027	-0.008	-0.051***	-0.023
	(0.019)	(0.018)	(0.018)	(0.019)
Black × N_{g1}	-0.029	-0.080***		
	(0.026)	(0.031)		
Male × N_{g1}			-0.037	0.004
			(0.031)	(0.026)
Constant	6.318***	-0.037	4.166**	2.877*
	(1.971)	(1.997)	(1.963)	(1.709)
Manager controls	Yes	Yes	Yes	Yes
Observations	481	318	409	390

Note: The column-4 Known-Effort specification of Table 4, re-estimated within worker subgroups (women, men, Black, White; Known-Effort arm, one observation per manager). In the gender-split columns, the worker's race and its interaction with observed effort N_{g1} are shown (the gender indicator is constant within column and omitted); in the race-split columns, the worker's gender and its interaction with effort are shown. *Black* × N_{g1} (*Male* × N_{g1}) is the differential effort slope of the sanction. All columns condition on the outcome and its interactions, observed effort, and the full set of manager controls. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Effort and Race: Intensive or Extensive Margins

	$\mathbf{1}[S > 0]$	$S \mid S > 0$	$\mathbf{1}[S = 10]$
Black worker	0.165**	1.308	0.080
	(0.083)	(0.942)	(0.056)
Black × Effort	-0.006**	-0.070*	-0.003*
	(0.003)	(0.036)	(0.002)
Effort (N_{g1})	-0.004	0.019	-0.000
	(0.002)	(0.032)	(0.002)
Male worker	0.015	0.589	0.023
	(0.082)	(0.903)	(0.058)
Good outcome (G_{g1})	-0.300***	-2.550***	-0.094***
	(0.046)	(0.945)	(0.023)
Constant	0.603***	4.120	0.177
	(0.209)	(2.771)	(0.141)
Mean dep. var.	0.315	5.377	0.076
Observations	799	252	799

Note: Decomposition of the Known-Effort-arm Round-1 sanction into three margins, one observation per manager. Column 1 is a linear probability model for any punishment ($\mathbf{1}[S_{g1} > 0]$); column 2 is OLS for the sanction amount conditional on punishing ($S \mid S > 0$); column 3 is a linear probability model for the maximum sanction ($\mathbf{1}[S_{g1} = 10]$). All columns use the fullest Known-Effort specification: worker race and gender, the outcome, observed effort N_{g1} and their interactions, plus manager controls. *Black worker* is the Black–White gap at zero effort and a bad outcome, for both genders; *Black × Effort* is how that gap changes with observed effort. *Mean dep. var.* is the outcome mean on the estimation sample. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: Main Results by Manager Gender

	Female manager	Male manager	p: equal
Panel A. Round-1 sanction S_{g1}, Known-Effort arm, all workers (N = 488, 311)			
Black worker	0.930	1.883*	0.447
	(0.723)	(1.031)	
Black × N_{g1}	-0.048*	-0.054	0.886
	(0.026)	(0.034)	
Male worker	0.251	-0.097	0.787
	(0.754)	(1.043)	
Male × N_{g1}	-0.009	0.007	0.710
	(0.026)	(0.034)	
Panel B. Round-1 sanction S_{g1}, Known-Effort arm, male workers (N = 181, 137)			
Black worker	2.529**	1.775	0.691
	(1.248)	(1.439)	
Black × N_{g1}	-0.110**	-0.038	0.263
	(0.045)	(0.046)	
Panel C. Round-2 worker effort (sliders), all workers (N = 967, 607)			
Punishment arm	0.731	1.219*	0.594
	(0.601)	(0.690)	
Pun. arm × Sanction	-0.468*	-0.316	0.709
	(0.240)	(0.329)	
Male worker	-0.939	-0.357	0.537
	(0.613)	(0.715)	
Black worker	0.771	1.881***	0.215
	(0.602)	(0.662)	
Panel D. Opts in to Round 3 with the same manager (0/1), all workers (N = 967, 607)			
Punishment arm	0.017	0.093**	0.169
	(0.034)	(0.043)	
Pun. arm × Sanction	-0.026**	-0.032**	0.757
	(0.011)	(0.013)	
Male worker	0.010	0.042	0.511
	(0.031)	(0.039)	
Black worker	0.021	0.011	0.843
	(0.031)	(0.039)	

Note: Key coefficients of the main specifications re-estimated separately for female and male managers (the manager's own gender). Panels A and B use the column-4 Known-Effort specification of Table 4 (all workers) and its male-worker version (Table A12), with the manager's gender dropped from the control set; Panel C uses column 5 of Table 6; Panel D uses column 1 of Table 8. *Black worker* is the Black–White gap in the sanction at zero effort and *Black* × N_{g1} its change per slider completed; *Pun. arm* × *Sanction* is the effect of an implemented sanction relative to the No-Punishment counterfactual. The last column is the p-value of a Wald test that the coefficient is equal across manager gender, from a pooled regression in which every regressor is interacted with the manager-gender indicator. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A15: Distribution of Managers' Beliefs and Perceived Effort Gaps (1)

Measure (sliders)	N	Mean	SD	p25	Median	p75
Manager: worker *should* do	1574	33.48	15.35	20	35	50
Manager: worker *will* do	1574	27.93	14.44	15	25	40
Expectations by worker race						
Should do, White worker	785	33.28	15.31	20	35	50
Should do, Black worker	789	33.68	15.40	20	35	50
Difference (Black − White)		0.40				
Will do, White worker	785	27.62	14.38	15	25	40
Will do, Black worker	789	28.24	14.49	18	30	40
Difference (Black − White)		0.62				
Expectations by worker gender						
Should do, Female worker	930	32.89	15.54	20	30	50
Should do, Male worker	644	34.32	15.05	24	35	50
Difference (Male − Female)		1.43*				
Will do, Female worker	930	27.99	14.44	15	30	40
Will do, Male worker	644	27.85	14.44	16	25	40
Difference (Male − Female)		-0.15				

Note: This table describes managers' Round-1 beliefs about worker effort, one observation per manager, all measured in sliders. It reports the manager's beliefs about how many sliders the worker *should* and *will* do, elicited from all managers before the outcome is realized, overall and separately by the worker's race and gender, with difference rows giving the Black−White and Male−Female gap in means; asterisks denote statistical significance from a heteroskedasticity-robust regression of the expectation on the single group indicator: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: Distribution of Managers' Beliefs and Perceived Effort Gaps (2)

Measure (sliders)	N	Mean	SD	p25	Median	p75
Manager's guess, Hidden Effort only ($\hat{N}_{g1}$)	775	22.17	14.19	10	20	30
if good outcome	270	29.37	15.45	15	30	45
if bad outcome	505	18.32	11.79	10	15	25
Guess by worker race						
Good outcome: White worker	149	27.72	15.62	15	25	45
Good outcome: Black worker	121	31.40	15.06	20	30	50
Good outcome: Difference (Black − White)		3.69*				
Bad outcome: White worker	246	18.03	11.54	10	15	25
Bad outcome: Black worker	259	18.59	12.03	10	20	27
Bad outcome: Difference (Black − White)		0.57				
Guess by worker gender						
Good outcome: Female worker	152	28.84	15.33	15	26	45
Good outcome: Male worker	118	30.05	15.65	15	30	45
Good outcome: Difference (Male − Female)		1.21				
Bad outcome: Female worker	297	17.92	11.35	10	15	25
Bad outcome: Male worker	208	18.88	12.39	10	17	28
Bad outcome: Difference (Male − Female)		0.96				

Note: This table describes managers' Round-1 beliefs about worker effort, one observation per manager, all measured in sliders. It reports the manager's guessed effort ($\hat{N}_{g1}$), overall, and separately by outcome, worker's race and gender. Because the guess is elicited only in the Hidden-Effort arm, these rows are restricted to that arm and so have roughly half as many observations. Difference rows giving the Black−White and Male−Female gap in means; asterisks denote statistical significance from a heteroskedasticity-robust regression of the expectation on the single group indicator: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: Expectations

	Hidden-effort arm: S_{g1}				
	(1)	(2)	(3)	(4)	(5)
Good outcome (G_{g1})	-1.653***	-1.292***	-1.297***	-1.423***	-1.412***
	(0.160)	(0.161)	(0.161)	(0.176)	(0.177)
Should $-\hat{N}_{g1}$		0.034***	0.022**		
		(0.008)	(0.010)		
Will $-\hat{N}_{g1}$				0.022**	0.007
				(0.010)	(0.011)
Black worker	-0.052	-0.020	-0.302	-0.026	-0.203
	(0.193)	(0.191)	(0.200)	(0.194)	(0.182)
Black × (Should $-\hat{N}_{g1}$)			0.025		
			(0.016)		
Black × (Will $-\hat{N}_{g1}$)					0.031*
					(0.018)
Male worker	0.082	0.067	0.053	0.122	0.096
	(0.196)	(0.193)	(0.192)	(0.197)	(0.196)
Punishment arm	0.113	0.112	0.117	0.127	0.115
	(0.193)	(0.190)	(0.189)	(0.192)	(0.192)
Constant	1.821***	1.304***	1.449***	1.575***	1.682***
	(0.207)	(0.227)	(0.222)	(0.232)	(0.226)
R^2	0.081	0.109	0.114	0.089	0.094
Observations	775	775	775	775	775

Note: OLS of the Round-1 sanction S_{g1} (tokens, 0–10) in the Hidden-Effort arm, where the manager does not observe effort and the benchmark is the manager's *guessed* effort $\hat{N}_{g1}$; one observation per manager. *Should* $-\hat{N}_{g1}$ is the normative shortfall (sliders the worker *should* do minus guessed effort) and *Will* $-\hat{N}_{g1}$ the expectational shortfall (sliders the worker *will* do minus guessed effort); a positive shortfall means the worker fell short and predicts more punishment. Column 1 is a baseline (outcome, worker gender, punishment arm); columns 2–3 add the normative shortfall and its interaction with worker race; columns 4–5 replace it with the expectational shortfall and its race interaction. The interactions test whether the shortfall-to-punishment slope differs for Black workers. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Expectations

	Known-effort arm: S_{g1}				
	(1)	(2)	(3)	(4)	(5)
Good outcome (G_{g1})	-1.778***	-1.647***	-1.650***	-1.753***	-1.754***
	(0.171)	(0.169)	(0.169)	(0.173)	(0.173)
Should $-$ N_{g1}		0.019***	0.012		
		(0.006)	(0.008)		
Will $-$ N_{g1}				0.004	0.001
				(0.006)	(0.008)
Black worker	0.433**	0.345*	0.240	0.412*	0.403*
	(0.207)	(0.207)	(0.215)	(0.210)	(0.211)
Black $\times$ (Should $-$ N_{g1})			0.013		
			(0.012)		
Black $\times$ (Will $-$ N_{g1})					0.005
					(0.012)
Male worker	0.120	0.094	0.096	0.118	0.116
	(0.214)	(0.212)	(0.212)	(0.214)	(0.214)
Punishment arm	0.026	0.039	0.029	0.027	0.023
	(0.208)	(0.207)	(0.206)	(0.208)	(0.208)
Constant	1.952***	1.815***	1.855***	1.947***	1.948***
	(0.227)	(0.229)	(0.227)	(0.228)	(0.227)
R^2	0.080	0.092	0.094	0.080	0.081
Observations	799	799	799	799	799

Note: OLS of the Round-1 sanction S_{g1} (tokens, 0–10) in the Known-Effort arm, where the manager observes the worker's actual effort N_{g1}; one observation per manager. *Should* $-$ N_{g1} is the normative shortfall (sliders the worker *should* do minus observed effort) and *Will* $-$ N_{g1} the expectational shortfall (sliders the worker *will* do minus observed effort); a positive shortfall means the worker fell short and predicts more punishment. Column 1 is a baseline (outcome, worker gender, punishment arm); columns 2–3 add the normative shortfall and its interaction with worker race; columns 4–5 replace it with the expectational shortfall and its race interaction. The interactions test whether the shortfall-to-punishment slope differs for Black workers. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Belief revision about Round-2 effort and the Round-1 sanction

	$\text{Will}_{g2} - \text{Will}_{g1}$			
	Continuous sanction		Punished dummy	
	No punishment	Punishment	No punishment	Punishment
Sanction (S_{g1})	0.027	-0.200		
	(0.644)	(0.453)		
Sanction × Black worker	0.338	0.239		
	(0.736)	(0.665)		
Punished ($S_{g1} > 0$)			-4.283	-2.180
			(3.492)	(3.438)
Punished × Black worker			6.756	0.239
			(4.859)	(4.887)
Black worker	-1.808	-5.480*	-3.213	-5.148
	(2.503)	(3.211)	(2.633)	(3.315)
Good outcome (G_{g1})	-0.780	3.043	-1.432	2.778
	(2.341)	(3.005)	(2.343)	(2.928)
Round-1 effort (N_{g1})	0.632***	0.589***	0.623***	0.572***
	(0.088)	(0.117)	(0.087)	(0.115)
Constant	-35.619**	-18.762	-33.459**	-18.053
	(14.525)	(13.548)	(14.530)	(13.480)
Observations	178	140	178	140
R^2	0.308	0.364	0.315	0.367

Note: OLS of the change in the manager's stated expectation of the worker's effort between Round 1 and Round 2, $\text{Will}_{g2} - \text{Will}_{g1}$ (sliders), on the Round-1 sanction and its interaction with worker race; restricted to the Known-Effort arm and to managers matched with a male worker, where the racial gap in sanctions is found; one observation per manager. Columns 1 and 2 use the sanction S_{g1} (tokens, 0–10); columns 3 and 4 use an indicator for a positive sanction ($S_{g1} > 0$). Each pair is estimated separately for the No-Punishment arm, where the chosen sanction is not implemented, and the Punishment arm, where it is. *Black worker* denotes the worker's avatar race (base: White). All columns condition on the Round-1 outcome G_{g1}, observed effort N_{g1}, and the full set of manager controls. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A20: Reasoning

Reason (LLM-induced)	N	% of answers
Effort evaluation	407	25.6
Fairness	251	15.8
Personal gain	227	14.3
Outcome-based	208	13.1
Motivation	121	7.6
Unwarranted	100	6.3
Empathy	99	6.2
Lack of control	94	5.9
Arbitrary	37	2.3
Trust	28	1.8
Punishment as incentive	20	1.3
All	1592	100.0

Note: Distribution of managers' Round-1 open-ended punishment rationales across categories generated bottom-up by a large language model (the model's own, data-driven taxonomy, distinct from the pre-specified categories used elsewhere). Each answer is assigned a single primary category, so the shares are mutually exclusive and sum to 100%. N is the number of answers in each category and % of answers is its share. One observation per manager's Round-1 rationale.

Table A21: Reasoning

	S_{g1}			
	(1)	(2)	(3)	(4)
Good outcome (G_{g1})	-1.747***	-1.976***	-1.962***	-2.014***
	(0.118)	(0.135)	(0.136)	(0.136)
Known effort arm	0.298**	-0.021	-0.026	0.004
	(0.143)	(0.144)	(0.144)	(0.143)
Punishment arm	0.079	0.050	0.054	0.069
	(0.144)	(0.137)	(0.138)	(0.135)
Effort evaluation		1.398***	1.398***	1.323***
		(0.237)	(0.237)	(0.239)
Fairness		0.905***	0.890***	0.811***
		(0.237)	(0.237)	(0.235)
Outcome-based		1.679***	1.682***	1.652***
		(0.274)	(0.275)	(0.271)
Empathy		-0.437*	-0.441*	-0.437*
		(0.235)	(0.237)	(0.248)
Lack of control		-0.721***	-0.707***	-0.765***
		(0.218)	(0.217)	(0.219)
Motivation/incentive		1.399***	1.405***	1.328***
		(0.319)	(0.318)	(0.312)
Unwarranted		-0.446**	-0.439**	-0.407*
		(0.210)	(0.210)	(0.218)
Arbitrary		3.451***	3.447***	3.032***
		(0.591)	(0.593)	(0.594)
Black worker			0.207	0.059
			(0.137)	(0.138)
Male worker			0.045	0.043
			(0.140)	(0.138)
Constant	1.903***	1.318***	1.193***	1.579*
	(0.137)	(0.187)	(0.209)	(0.933)
Categories joint p		0.000	0.000	0.000
Manager controls	No	No	No	Yes
R^2	0.082	0.173	0.175	0.213
Observations	1,546	1,546	1,546	1,546

Note: OLS of the Round-1 sanction S_{g1} (tokens, 0–10) on indicators for the manager's (LLM-classified, bottom-up) reasoning category, one observation per manager, restricted to the seven sizable categories. Column 1 is a baseline (outcome and treatment-arm indicators); column 2 adds the reasoning-category fixed effects (base category: *Personal gain*), so each coefficient is the mean sanction relative to that group; column 3 adds worker race and gender; column 4 adds manager controls. *Categories joint p* is an F-test that the categories differ. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A22: Reasoning

	Effort	Fair.	Outc.	Gain	Empat.	Control	Motiv/inc	Unwarr.	Arbit.
Known effort arm	0.247***	-0.068***	-0.047***	-0.058***	-0.039***	-0.041***	0.027*	-0.013	0.011
	(0.021)	(0.018)	(0.016)	(0.018)	(0.012)	(0.012)	(0.014)	(0.012)	(0.008)
Black worker	-0.010	0.040**	-0.011	0.006	0.005	-0.018	-0.014	-0.005	-0.006
	(0.022)	(0.019)	(0.017)	(0.018)	(0.012)	(0.012)	(0.014)	(0.013)	(0.008)
Male worker	0.022	-0.000	0.002	-0.019	0.001	-0.005	-0.008	-0.006	0.005
	(0.022)	(0.018)	(0.017)	(0.018)	(0.013)	(0.012)	(0.015)	(0.012)	(0.008)
Mean share (%)	25.7	15.8	12.8	14.2	6.2	6.0	8.8	6.4	2.4
Observations	1,574	1,574	1,574	1,574	1,574	1,574	1,574	1,574	1,574

Note: Linear probability models for the manager's use of each (LLM-classified, bottom-up) reasoning category; one regression per column, the dependent variable being a 0/1 indicator for that category, one observation per manager. Reported are the coefficients on the Known-Effort arm and on worker race and gender, each net of the others, the outcome, the punishment arm, and manager controls; a coefficient is the change in the probability of invoking that category. *Mean share* is the base rate of the category. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A23: Racial gap in Round-1 sanctions by managers' stated reason

Stated reason	N	Gap at zero effort	Gap at 10 sliders	Black × Effort
Effort evaluation	130	3.914***	2.431**	-0.148***
		(1.264)	(0.946)	(0.041)
Outcome	29	4.423**	3.378**	-0.105
		(2.103)	(1.595)	(0.073)
Personal gain	32	0.548	0.570	0.002
		(1.676)	(1.136)	(0.065)
Empathy	17	-5.572*	-3.679*	0.189*
		(2.851)	(1.912)	(0.108)
Motivation	32	0.198	0.617	0.042
		(3.373)	(2.282)	(0.119)
Unwarranted	20	1.539	0.899	-0.064
		(1.349)	(0.968)	(0.044)
Fairness	34	-1.206	0.171	0.138
		(2.044)	(1.362)	(0.085)
p-value: equal across all reasons		0.038		0.007
Other (categories with fewer than 15 managers, pooled)	24		estimated, not shown	
Observations	318			

Note: OLS of the Round-1 sanction S_{g1} (tokens, 0–10) on indicators for the manager's (LLM-classified, bottom-up) reasoning category, each interacted with the Black-worker indicator and with observed effort N_{g1}; restricted to the Known-Effort arm and to managers matched with a male worker, where the racial gap in sanctions is found; one observation per manager. Each row reports, for managers in that category, the Black–White difference in the sanction at zero effort, at 10 sliders, and its change per additional slider, computed from one regression that also conditions on the Round-1 outcome G_{g1} and its interaction with race, the punishment arm, and the full set of manager controls. The two p-value rows are Wald tests that the gap at zero effort (column 3) or the effort slope (column 5) is equal across all categories, and across effort evaluation, outcome, and arbitrary only. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A24: Racial gap in Round-1 sanctions: performance reasons versus other reasons

	N	Gap at zero effort	Gap at 10 sliders	Black × Effort
Performance reasons	191	3.730***	2.458***	-0.127***
		(1.101)	(0.820)	(0.036)
Other reasons	127	0.230	0.313	0.008
		(1.124)	(0.760)	(0.044)
Difference		3.500**	2.145*	-0.135**
		(1.606)	(1.096)	(0.059)
Observations	318			

Note: OLS of the Round-1 sanction S_{g1} (tokens, 0–10) on an indicator for a performance-based reason, interacted with the Black-worker indicator and with observed effort N_{g1}; restricted to the Known-Effort arm and to managers matched with a male worker; one observation per manager. *Performance reasons* are the (LLM-classified, bottom-up) categories in which the manager judges the worker's performance, past or expected: effort evaluation, outcome, and motivation. *Other reasons* are fairness, empathy, personal gain, lack of control, unwarranted (no reason to fine), and arbitrary (no reason for the amount). Each row reports, for managers in that group, the Black–White difference in the sanction at zero effort, at 10 sliders, and its change per additional slider; *Difference* is the performance-minus-other gap in each column. All quantities come from one regression that also conditions on the Round-1 outcome G_{g1} and its interaction with race, the punishment arm, and the full set of manager controls. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A25: Word-list check on the justifications for low-effort workers, by worker race

	White worker	Black worker	Difference	p-value
All *workers with 26 or fewer sliders (N = 70 White, 100 Black)*				
Blame / character words	0.157	0.230	+0.073	0.242
Encouragement / forward-looking words	0.200	0.100	-0.100*	0.065
Softening / excuse words	0.329	0.280	-0.049	0.496
Fined *workers with 26 or fewer sliders (N = 23 White, 43 Black)*				
Blame / character words	0.130	0.256	+0.125	0.235
Encouragement / forward-looking words	0.304	0.186	-0.118	0.274
Softening / excuse words	0.391	0.186	-0.205*	0.069
All *workers with 12 or fewer sliders (N = 17 White, 23 Black)*				
Blame / character words	0.235	0.348	+0.113	0.443
Encouragement / forward-looking words	0.176	0.043	-0.133	0.166
Softening / excuse words	0.529	0.261	-0.269*	0.083
Fined *workers with 12 or fewer sliders (N = 6 White, 15 Black)*				
Blame / character words	0.333	0.467	+0.133	0.577
Encouragement / forward-looking words	0.333	0.067	-0.267	0.115
Softening / excuse words	0.667	0.067	-0.600***	0.004

Note: Share of managers' written Round-1 justifications that contain at least one word from a fixed list, by worker race; restricted to the Known-Effort arm and to managers matched with a male worker, one observation per manager. The three lists correspond to the explanations of the gap considered in the text: *Blame/character* (lazy, nothing, unacceptable, deserve, didn't try, no effort, bare minimum, didn't care, slack, consequence), *Encouragement/forward-looking* (encourage, motivate, hope, next round, try harder, do better, improve, lesson), and *Softening/excuse* (fair, first try, not his fault, didn't want to, felt bad, understand, he tried, did his best). Justifications are lower-cased and matched at word boundaries, with stems where a word has several forms. Panels restrict the sample by the worker's Round-1 effort (26 or fewer sliders, the sample mean; 12 or fewer) and by whether the manager imposed a positive sanction (*Fined*). *Difference* is the Black-minus-White difference in shares; the p-value is from a two-sample test of proportions. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A26: Low-effort penalty in the Known-Effort arm, Round 1 vs. Round 2

	S_{gt}, KE arm	
	Round 1	Round 2
Black worker	1.625*	-0.037
	(0.855)	(1.001)
Male worker	0.280	-0.307
	(0.897)	(1.048)
Good outcome	-1.873***	-1.023**
	(0.437)	(0.401)
Black × Good outcome	0.231	0.410
	(0.520)	(0.560)
Effort (N_{gt})	-0.013	-0.125***
	(0.026)	(0.022)
Black × Effort	-0.060*	-0.006
	(0.031)	(0.029)
Male × Effort	-0.006	0.013
	(0.032)	(0.029)
Constant	3.360	5.676***
	(2.086)	(1.924)
Observations	389	389

Note: OLS of the Round-t sanction on worker race and gender, the outcome, observed effort, and their interactions, with manager controls; Known-Effort ∩ Punishment arm. *Black worker* is the low-effort ($N = 0$, bad outcome) racial gap; *Black* × *Effort* is the differential effort slope. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A27: Do managers credit effort improvements equally by race?

	Δ Sanction (R2−R1)	
	(1)	(2)
Δ Effort (R2−R1)	-0.084***	-0.130***
	(0.022)	(0.033)
Black worker	0.340	0.331
	(0.427)	(0.399)
Black × Δ Effort	-0.077	-0.075*
	(0.047)	(0.044)
Male worker		-0.832*
		(0.426)
Constant	0.418*	-0.683
	(0.246)	(2.313)
Manager controls	No	Yes
Observations	389	389

Note: OLS of the change in the sanction (Round 2 − Round 1) on the change in observed effort and its interaction with worker race; Known-Effort ∩ Punishment arm. A negative Δ *Effort* coefficient means managers lower the sanction when the worker improves; *Black* × Δ *Effort* tests whether managers credit improvement differently for Black workers. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A28: Manager's Reasoning and Traits and Differential Punishment

	Round-1 sanction (tokens)		
	Performance reasons	Traits	Both
Black worker	0.230	2.353**	0.310
	(1.124)	(0.950)	(1.173)
Black × Effort (N_{g1})	0.008	-0.078**	0.007
	(0.044)	(0.032)	(0.045)
Black × Performance reasoner	3.500**		3.264**
	(1.606)		(1.639)
Black × Performance reasoner × Effort	-0.135**		-0.130**
	(0.059)		(0.059)
Black × Negative reciprocity (z)		0.601*	0.691**
		(0.333)	(0.315)
Black × Altruism (z)		-0.058	-0.121
		(0.348)	(0.340)
Performance reasoner	2.166*		2.436**
	(1.164)		(1.211)
Performance reasoner × Effort	-0.044		-0.050
	(0.039)		(0.039)
Negative reciprocity (z)		-0.118	-0.199
		(0.208)	(0.211)
Altruism (z)		0.302	0.376
		(0.268)	(0.265)
Effort (N_{g1})	0.021	-0.011	0.023
	(0.029)	(0.018)	(0.030)
Good outcome (G_{g1})	-1.593***	-1.567***	-1.576***
	(0.396)	(0.385)	(0.394)
Observations	318	318	318
R^2	0.298	0.228	0.310

Note: OLS of the Round-1 sanction S_{g1} (tokens, 0–10); restricted to the Known-Effort arm and to managers matched with a male worker, where the racial gap in sanctions is found; one observation per manager. Column 1 interacts the Black-worker indicator and observed effort N_{g1} with an indicator for a performance-based stated reason (effort evaluation, outcome, or motivation, from the LLM classification of managers' explanations). Column 2 instead interacts the Black-worker indicator with two manager traits from the survey, negative reciprocity and altruism, each standardized within the sample. Column 3 includes both sets of interactions. All columns condition on the Round-1 outcome G_{g1} and its interaction with race, the punishment arm, and the full set of manager controls, with the raw altruism and negative-reciprocity scores replaced by their standardized versions in columns 2 and 3. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A29: Manager's Traits and Differential Punishment

	Sanction (S_{g1})	Punished ($\mathbf{1}[S > 0]$)
Black worker	0.157	0.018
	(0.140)	(0.022)
Black × Male (z)	0.022	-0.007
	(0.143)	(0.022)
Black × Age (z)	-0.294**	-0.017
	(0.145)	(0.023)
Black × H.S. or less (z)	-0.001	-0.008
	(0.140)	(0.022)
Black × Employed (z)	-0.044	0.005
	(0.139)	(0.022)
Black × Risk-taking (z)	0.059	-0.002
	(0.152)	(0.023)
Black × Time disc. (z)	-0.198	-0.033
	(0.156)	(0.023)
Black × Altruism (z)	0.435***	0.049**
	(0.154)	(0.023)
Black × Trust (z)	-0.146	0.003
	(0.220)	(0.033)
Black × Best intentions (z)	0.037	-0.023
	(0.207)	(0.033)
Black × Pos. reciprocity (z)	-0.079	0.011
	(0.153)	(0.023)
Black × Neg. reciprocity (z)	0.254	0.055**
	(0.159)	(0.024)
Black × Effort cost per slider (z)	0.003	0.012
	(0.142)	(0.021)
Controls	Yes	Yes
Observations	1,574	1,574

Manager characteristics z-scored. Controls: outcome, KE arm, outcome×KE.

Note: OLS (column 1, sanction S_{g1} in tokens) and linear probability (column 2, any punishment $\mathbf{1}[S_{g1} > 0]$) of the Round-1 decision on worker race interacted *jointly* with twelve manager characteristics, one observation per manager, all arms pooled. Each manager characteristic is standardized (mean 0, SD 1); the effort cost per slider is standardized within the price list the manager saw (15, 30, or 45 sliders), because the same wage offers imply a mechanically lower per-slider cost on longer lists. The reported *Black × characteristic* coefficients show the shift in the Black–White gap per one-standard-deviation increase in that characteristic, all entered simultaneously to show which moderators survive mutual adjustment. *Black worker* is the gap at the average manager. All models control for the outcome, the Known-Effort arm, their interaction, and the twelve characteristic main effects (not shown). Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 3: Distribution of Sliders Completed in Round 1

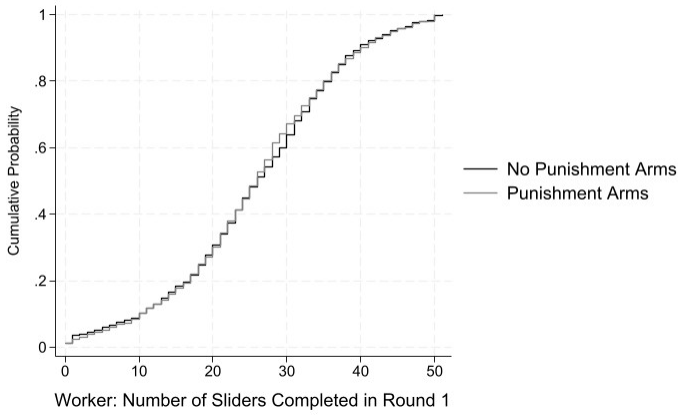


Table A30: Predictors of Effort in Round 1

Sliders Completed in Round 1			
	All	No punishment	Punishment
Black worker	-5.28***	-4.83***	-5.61***
	(0.58)	(0.84)	(0.82)
Male worker	0.89	0.49	1.43*
	(0.55)	(0.77)	(0.80)
Age	-0.20***	-0.21***	-0.19***
	(0.021)	(0.030)	(0.031)
High school	4.01	10.1*	-0.13
	(3.65)	(5.17)	(4.21)
Trade/vocational	1.76	7.61	-2.47
	(3.71)	(5.25)	(4.36)
Associate (in prog.)	4.86	8.43	3.72
	(3.91)	(5.45)	(4.80)
Associate degree	1.87	6.96	-1.47
	(3.68)	(5.23)	(4.24)
Bachelor's (in prog.)	4.83	11.9**	-0.33
	(3.79)	(5.29)	(4.48)
Bachelor's degree	3.28	8.77*	-0.57
	(3.63)	(5.12)	(4.16)
Advanced (in prog.)	1.98	6.42	-0.63
	(3.79)	(5.39)	(4.36)
Advanced degree	0.57	5.64	-2.84
	(3.66)	(5.17)	(4.21)
Job seeking	3.81***	2.83*	4.25***
	(1.10)	(1.57)	(1.54)
Employed	0.52	0.26	0.54
	(0.79)	(1.11)	(1.10)
Male manager avatar	0.72	1.29*	0.16
	(0.55)	(0.77)	(0.77)
Effort cost	-0.076	-0.33	0.10
	(0.36)	(0.50)	(0.51)
Risk-taking	-1.32***	-0.83*	-1.86***
	(0.32)	(0.47)	(0.43)
Patience	0.47	0.46	0.42
	(0.37)	(0.52)	(0.52)
Altruism	-0.17	0.43	-0.96*
	(0.40)	(0.60)	(0.52)
Trust	-0.51	-0.35	-0.69
	(0.39)	(0.50)	(0.60)
Good intentions	-0.45	-0.41	-0.58
	(0.39)	(0.49)	(0.61)
Positive reciprocity	1.77***	1.48**	2.37***
	(0.47)	(0.65)	(0.66)
Negative reciprocity	-0.58**	-1.12***	-0.11
	(0.28)	(0.39)	(0.39)

Table A31: Round 2 effort responses to punishment, by Round-1 effort

	Round 2 Effort (Sliders Completed)		
	R1 effort ≤ 26	R1 effort > 26	Pooled
Punishment Arm	0.52	1.29**	0.54
	(0.66)	(0.63)	(0.66)
R1 Sanction Amt.	0.11	-0.61**	0.08
	(0.13)	(0.25)	(0.13)
Pun. Arm × Sanction Amt.	-0.34*	-0.43	-0.34*
	(0.19)	(0.39)	(0.19)
R1 effort above median			1.78*
			(0.97)
Above median × Pun. Arm			0.76
			(0.91)
Above median × Sanction Amt.			-0.66**
			(0.28)
Above median × Pun. Arm × Sanction Amt.			-0.12
			(0.44)
R1 effort	0.83***	0.70***	0.77***
	(0.05)	(0.06)	(0.04)
Good outcome	-0.00	-0.16	-0.09
	(0.66)	(0.65)	(0.46)
Known-effort arm	0.44	1.99***	1.22***
	(0.60)	(0.67)	(0.45)
Constant	8.17***	12.86***	8.77***
	(0.98)	(2.27)	(0.84)
Observations	816	758	1,574
Adjusted R^2	0.339	0.235	0.536

Note: OLS of Round 2 worker effort N_{g2} (sliders), one observation per group, using the specification of Column 5 of Table 6: the Punishment-arm indicator, the manager's Round 1 sanction, and their interaction, conditioning on Round 1 effort, the Round 1 outcome, and the Known-Effort arm. Columns 1 and 2 split the sample at the median of Round 1 effort (26 sliders). Column 3 pools both. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A32: Responses to Punishment by Race and Gender

	Round 2 Effort (Sliders Completed)
Punishment Arm	1.41*
	(0.83)
R1 Sanction Amt.	-0.39*
	(0.22)
Pun. Arm × Sanction Amt.	-0.34
	(0.32)
R1 effort	0.82***
	(0.023)
Good outcome	0.018
	(0.46)
Known-effort arm	1.26***
	(0.45)
Male worker	-1.46**
	(0.70)
Black worker	1.26*
	(0.69)
Black × Sanction Amt.	0.14
	(0.24)
Male × Sanction Amt.	0.33
	(0.25)
White × Pun. Arm	0.93
	(0.91)
Female × Pun. Arm	-1.65*
	(0.95)
Black × Pun. Arm × Sanction Amt.	0.38
	(0.41)
Male × Pun. Arm × Sanction Amt.	-0.71*
	(0.41)
Constant	8.22***
	(0.93)
Observations	1,574
Adjusted R^2	0.534

Note: OLS of Round 2 worker effort N_{g2} (sliders) pooled across all workers. All columns use heteroskedasticity-robust standard errors; standard errors are displayed in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A33: Group Efficiency (R1 - R2)

	Joint surplus	Wkr pay	Mgr pay	Mgr share	Welfare
Punishment arm	-1.312***	-1.547***	0.235	0.034***	-1.194***
	(0.304)	(0.136)	(0.247)	(0.013)	(0.328)
Known effort arm	-0.011	-0.000	-0.011	-0.001	0.076
	(0.243)	(0.000)	(0.243)	(0.012)	(0.271)
Punishment × Known effort	-0.431	-0.143	-0.288	-0.016	-0.658
	(0.428)	(0.192)	(0.346)	(0.018)	(0.455)
Round 2 FE	0.113	-0.090	0.203	0.014*	-0.221
	(0.185)	(0.059)	(0.164)	(0.009)	(0.184)
Constant	13.260***	10.045***	3.215***	0.159***	10.866***
	(0.198)	(0.029)	(0.194)	(0.010)	(0.216)
R^2	0.019	0.120	0.001	0.004	0.019
Group-rounds	3,148	3,148	3,148	3,016	3,148

Note: OLS on the pooled Rounds 1–2 group-round panel (one observation per group per round, all arms). Dependent variables: *Joint surplus* (manager plus worker payoff, tokens), *Worker pay*, *Manager pay*, *Manager share* (manager's share of realized group earnings, defined when surplus is positive), and *Welfare* (surplus net of the worker's effort cost). Regressors are the two treatment arms and their interaction (base: No-punishment × Hidden-effort) and a Round-2 fixed effect. Standard errors are clustered by group in parentheses. One token = \$0.10. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A34: Group Efficiency (R1 - R3)

	Joint surplus	Wkr pay	Mgr pay	Mgr share	Welfare
Punishment arm	-1.229***	-1.553***	0.324	0.040***	-1.057***
	(0.283)	(0.131)	(0.228)	(0.012)	(0.310)
Known effort arm	0.185	0.000	0.185	0.009	0.356
	(0.217)	(0.000)	(0.217)	(0.011)	(0.247)
Punishment × Known effort	-0.625	-0.133	-0.492	-0.026	-0.944**
	(0.398)	(0.188)	(0.314)	(0.017)	(0.427)
Round 2 FE	0.113	-0.090	0.203	0.014*	-0.221
	(0.185)	(0.059)	(0.164)	(0.009)	(0.184)
Round 3 FE	0.418*	-0.047	0.465**	0.028***	0.072
	(0.214)	(0.075)	(0.186)	(0.010)	(0.215)
Constant	13.167***	10.046***	3.122***	0.153***	10.728***
	(0.189)	(0.034)	(0.183)	(0.009)	(0.209)
R^2	0.020	0.118	0.002	0.006	0.019
Group-rounds	4,157	4,157	4,157	3,974	4,157

Note: As Table A33, estimated on the Rounds 1–3 group-round panel with Round-2 and Round-3 fixed effects. Round-3 observations are limited to the groups that continued to Round 3 (a selected subsample), so the Round-3 rows condition on continuation. Dependent variables and regressors are otherwise identical; standard errors are clustered by group in parentheses. One token = \$0.10. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A35: Group Efficiency: Cumulative (R1-R3)

	Joint surplus	Wkr pay	Mgr pay	Welfare
Punishment arm	-2.587***	-3.888***	1.301	-1.750
	(0.987)	(0.439)	(0.824)	(1.088)
Known effort arm	0.543	-0.000	0.543	1.645*
	(0.780)	(.)	(0.780)	(0.884)
Punishment × Known effort	-2.186	-0.371	-1.815	-3.730**
	(1.398)	(0.639)	(1.129)	(1.505)
Constant	40.442***	30.000	10.442***	32.350***
	(0.564)	(.)	(0.564)	(0.651)
R^2	0.030	0.144	0.003	0.029
Groups	1,009	1,009	1,009	1,009

Note: OLS across groups (one observation per group), restricted to groups that completed all three rounds. Dependent variables are per-group *cumulative* totals over Rounds 1–3: *Joint surplus*, *Worker pay*, *Manager pay*, and *Welfare* (surplus net of worker effort cost), in tokens. Regressors are the two treatment arms and their interaction (base: No-punishment × Hidden-effort). Heteroskedasticity-robust standard errors in parentheses. One token = \$0.10. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A36: Group Efficiency, by worker's characteristics (R1–R2 & R1–R3)

	R1–R2				R1–R3			
	Women	Men	Black	White	Women	Men	Black	White
Joint surplus	-1.413***	-1.672***	-1.791***	-1.285***	-1.358***	-1.798***	-1.842***	-1.262***
	(0.273)	(0.345)	(0.303)	(0.299)	(0.253)	(0.322)	(0.286)	(0.275)
Worker pay	-1.531***	-1.757***	-1.843***	-1.400***	-1.544***	-1.739***	-1.838***	-1.408***
	(0.118)	(0.162)	(0.146)	(0.125)	(0.116)	(0.159)	(0.141)	(0.124)
Manager pay	0.118	0.085	0.052	0.115	0.186	-0.059	-0.004	0.146
	(0.226)	(0.271)	(0.239)	(0.249)	(0.203)	(0.248)	(0.219)	(0.224)
Welfare	-1.385***	-1.707***	-1.652***	-1.410***	-1.332***	-1.821***	-1.722***	-1.353***
	(0.294)	(0.363)	(0.321)	(0.322)	(0.274)	(0.345)	(0.305)	(0.300)
Manager share	0.025**	0.030**	0.027**	0.024*	0.030***	0.023*	0.025**	0.028**
	(0.012)	(0.014)	(0.013)	(0.013)	(0.011)	(0.013)	(0.012)	(0.012)

Note: Each cell reports the coefficient on the Punishment arm from a separate regression of the group outcome (rows: joint surplus, worker pay, manager pay, welfare, and manager share) on the punishment and known-effort arm indicators and round fixed effects, estimated within the worker subgroup heading the column (Women, Men, Black, White). The two panels use the Rounds 1–2 panel and the Rounds 1–3 panel (Round-3 completers only). Standard errors clustered by group in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A37: Group Efficiency: distribution

Treatment arm	N	Mean	SD	% zero	% max
No punishment × Hidden effort	784	13.3	4.7	0.0	33.2
No punishment × Known effort	820	13.3	4.7	0.0	33.0
Punishment × Hidden effort	766	12.0	6.4	8.6	33.2
Punishment × Known effort	778	11.6	6.3	8.5	28.9
All	3148	12.6	5.6	4.2	32.1

Note: Descriptive distribution of joint group surplus (tokens) on the Rounds 1–2 group-round panel, by treatment arm. N is the number of group-rounds; % zero and % max are the shares with zero surplus (a bad outcome with no offsetting) and with the maximum surplus of 20 (a good outcome with no fine implemented), respectively. One token = \$0.10.

Table A38: Punishment and effort

	S_{g1}	
	(1)	(2)
Effort (N_{g1})	-0.006	0.011
	(0.009)	(0.009)
Effort × Known effort	-0.055***	-0.058***
	(0.013)	(0.013)
Known effort arm	1.811***	1.795***
	(0.395)	(0.386)
Punishment arm	0.042	0.058
	(0.145)	(0.140)
Good outcome		-1.633***
		(0.119)
Constant	1.450***	1.576***
	(0.263)	(0.255)
R^2	0.033	0.098
Observations	1,574	1,574

Note: OLS of the Round-1 sanction S_{g1} (tokens, 0–10) across groups (one observation per group, all arms). The key regressor is the interaction *Effort* × *Known effort*, which tests whether the sanction declines more steeply with the worker's effort N_{g1} when the manager observes it (base: Hidden-effort arm). Column 1 is the baseline; column 2 adds a good-outcome control. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A39: Effort Provision by treatment arm

	Round-2 effort N_{g2}		
	All groups	No punishment	Punishment
Known effort arm	1.278***	1.676***	0.864
	(0.450)	(0.640)	(0.632)
Round-1 effort	0.813***	0.833***	0.791***
	(0.024)	(0.028)	(0.039)
Round-1 sanction	-0.580***		-0.586***
	(0.155)		(0.158)
Punishment arm	1.176***		
	(0.440)		
Constant	8.261***	7.535***	10.214***
	(0.712)	(0.814)	(1.113)
R^2	0.531	0.530	0.534
Observations	1,574	802	772

Note: OLS of Round-2 worker effort N_{g2} (sliders) across groups (one observation per group). The key regressor is the *Known-effort arm* indicator, capturing the effect of the manager having observed effort in Round 1 on the worker's next-round effort, controlling for Round-1 effort and the Round-1 implemented sanction. Column 1 uses all groups; columns 2 and 3 split by whether punishment was possible (No-punishment vs. Punishment arm). Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A40: Group Efficiency and Observing Effort

	Effort	Joint surplus	Welfare
A. Punishment×Known-effort by round — full sample (Rounds 1–2)			
Round 1	-0.07	-0.41	-0.69
	(1.14)	(0.56)	(0.57)
Round 2	-1.06	-0.45	-0.62
	(1.30)	(0.58)	(0.60)
B. Punishment×Known-effort by round — Round-3 completers			
Round 1	-0.89	-0.25	-0.65
	(1.41)	(0.68)	(0.70)
Round 2	-0.52	-0.70	-1.24*
	(1.51)	(0.71)	(0.73)
Round 3	-0.62	-1.23*	-1.84**
	(1.56)	(0.73)	(0.75)
C. Difference across rounds (completers, triple interaction)			
Round 3 − Round 1	0.27	-0.98	-1.19
	(1.17)	(0.91)	(0.91)
Round 3 − Round 2	-0.10	-0.53	-0.60
	(0.92)	(0.91)	(0.91)

Note: Each cell reports the *Punishment* × *Known-effort* interaction coefficient from a separate regression of the column outcome—worker effort (sliders), joint surplus (tokens), and welfare (surplus net of effort cost, tokens). Panel A estimates it round by round on the full Rounds 1–2 sample; Panel B does so for the groups that reach Round 3 (Round 3 is played only by these continuers); Panel C reports, from a round × punishment × known-effort triple-interaction model, how the Round-3 interaction differs from Rounds 1 and 2. Panels A–B use heteroskedasticity-robust standard errors; Panel C clusters by group. Standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Robustness: No Failed Attention Checks

Table A41: Robustness: Punishment Decisions and Effort, Participants with No Failed Attention Check

	(1)	(2)	(3)	(4)
Black worker	0.147	-0.123	1.249	0.978
	(0.362)	(0.347)	(0.791)	(0.765)
Male worker	-0.109	-0.101	0.699	0.732
	(0.371)	(0.353)	(0.792)	(0.748)
Good outcome	-1.887***	-1.636***	-1.785***	-1.788***
	(0.330)	(0.326)	(0.322)	(0.328)
Black × Good outcome	-0.007	-0.009	0.198	0.085
	(0.411)	(0.405)	(0.389)	(0.400)
Male × Good outcome	0.376	0.357	0.486	0.501
	(0.425)	(0.422)	(0.397)	(0.408)
Effort (sliders)		-0.062***	-0.024	-0.023
		(0.012)	(0.022)	(0.021)
Black × Effort			-0.056**	-0.051**
			(0.025)	(0.024)
Male × Effort			-0.032	-0.034
			(0.025)	(0.023)
Constant	2.037***	3.707***	2.694***	5.278***
	(0.313)	(0.463)	(0.716)	(1.784)
Manager controls	No	No	No	Yes
Observations	488	488	488	488

Note: Table 4 re-estimated on the sample of participants who failed no attention check, the exclusion rule in the original pre-registration (see Section 2). OLS of the Round-1 sanction S_{g1} (tokens, 0–10), Known-Effort arm, one observation per manager. Column 1 conditions on the outcome and its race and gender interactions; column 2 adds observed effort N_{g1}; column 3 adds *Black/Male* × N_{g1}; column 4 adds manager controls (as in the main text, minus the attention-check count, which is zero for all managers here). Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A42: Robustness: Effort, Gender and Race, Participants with No Failed Attention Check

	(1) Women	(2) Men	(3) Black	(4) White
Black worker	-0.543	3.316***		
	(1.010)	(1.204)		
Male worker			2.631**	-1.036
			(1.120)	(1.066)
Good outcome	-1.840***	-1.091***	-1.585***	-1.933***
	(0.371)	(0.407)	(0.379)	(0.369)
Black × Good outcome	0.296	-0.225		
	(0.521)	(0.685)		
Male × Good outcome			0.114	0.739
			(0.653)	(0.543)
Effort (sliders)	-0.047*	-0.036*	-0.050**	-0.043*
	(0.025)	(0.019)	(0.020)	(0.025)
Black × Effort	-0.006	-0.117***		
	(0.033)	(0.035)		
Male × Effort			-0.092**	0.010
			(0.036)	(0.031)
Constant	6.483**	5.112**	5.503**	4.748**
	(2.678)	(2.065)	(2.253)	(2.156)
Observations	293	195	238	250

Note: Table A12 re-estimated on the sample of participants who failed no attention check. The column-4 Known-Effort specification of Table A41, estimated within worker subgroups (women, men, Black, White; one observation per manager). In the gender-split columns, the worker's race and its interaction with observed effort N_{g1} are shown; in the race-split columns, the worker's gender and its interaction with effort are shown. All columns condition on the outcome and its interactions, observed effort, and manager controls (minus the attention-check count). Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A43: Robustness: Responses to Punishment by Race and Gender, Participants with No Failed Attention Check

	(1) White	(2) Black	(3) Men	(4) Women
Punishment arm	0.728	0.505	1.852*	-0.246
	(0.940)	(0.874)	(1.086)	(0.810)
R1 sanction	-0.147	-0.093	-0.031	-0.222
	(0.237)	(0.176)	(0.236)	(0.184)
Punishment × sanction	-1.022***	-0.184	-0.939***	-0.179
	(0.349)	(0.247)	(0.338)	(0.263)
R1 effort	0.819***	0.856***	0.830***	0.820***
	(0.038)	(0.038)	(0.042)	(0.036)
Good outcome	-0.636	1.226	-0.211	0.580
	(0.951)	(0.876)	(1.070)	(0.823)
Known-effort arm	1.015	0.279	0.520	0.983
	(0.857)	(0.773)	(0.973)	(0.724)
Male worker	-0.622	0.742		
	(0.878)	(0.778)		
Black worker			1.415	-0.260
			(1.014)	(0.738)
Constant	9.158***	7.417***	7.530***	8.904***
	(1.358)	(1.223)	(1.595)	(1.256)
Observations	503	445	375	573

Note: Table 7 re-estimated on the sample of workers who failed no attention check. OLS of Round 2 worker effort N_{g2} (sliders), one observation per group. Columns 1 and 2 present the results for white and Black workers, respectively; columns 3 and 4 for men and women. All columns control for Round 1 effort, the Round 1 outcome, and the Known-Effort arm. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A44: Robustness: Willingness to Continue with the Same Manager, Participants with No Failed Attention Check

	(1) All	(2) White	(3) Black	(4) Men	(5) Women
Punishment arm	0.058*	0.048	0.073	0.038	0.084*
	(0.035)	(0.046)	(0.053)	(0.055)	(0.045)
R1 sanction	0.001	-0.001	0.003	-0.005	0.007
	(0.008)	(0.012)	(0.011)	(0.012)	(0.010)
Punishment × sanction	-0.020*	-0.024	-0.019	-0.004	-0.032**
	(0.011)	(0.017)	(0.015)	(0.017)	(0.015)
R1 effort	0.006***	0.006***	0.006**	0.004**	0.008***
	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
Good outcome	0.032	0.060	-0.007	0.117**	-0.031
	(0.035)	(0.047)	(0.053)	(0.054)	(0.046)
Known-effort arm	0.002	0.005	0.000	0.028	-0.021
	(0.031)	(0.042)	(0.047)	(0.049)	(0.040)
Male worker		-0.001	0.071		
		(0.043)	(0.047)		
Black worker				0.056	-0.005
				(0.051)	(0.041)
Constant	0.443***	0.421***	0.432***	0.443***	0.400***
	(0.047)	(0.067)	(0.074)	(0.081)	(0.070)
Observations	948	503	445	375	573

Note: Table 8 re-estimated on the sample of workers who failed no attention check. Linear probability models of the worker's decision to opt in to Round 3 with the same manager, one observation per group. Column 1 presents all workers; columns 2 and 3 present the results for white and Black workers, respectively; columns 4 and 5 for men and women. All columns control for Round 1 effort, the Round 1 outcome, and the Known-Effort arm. Heteroskedasticity-robust standard errors in parentheses. Asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.